Organic Chemistry Booklet

Midterm 2 · Lectures 10 – 12

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10 Aldehydes & Ketones

In this lecture

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10.1 Structure of Aldehydes & Ketones

Q1 What is the difference between aldehydes and ketones?

A. Aldehydes have at least one H on the carbonyl carbon; ketones have two carbon groups on the carbonyl carbon

B. Aldehydes have a C=O in the middle of the chain; ketones have it at the end

C. Aldehydes contain an O–H group; ketones contain a C=O group

D. Aldehydes have two carbon groups on the carbonyl carbon; ketones have at least one H

Q2 Both aldehydes and ketones contain which functional group?

A. The carboxyl group (–COOH)

B. The ether group (C–O–C)

C. The carbonyl group (C=O)

D. The hydroxyl group (–OH)

Q3 CH_3CH_2CHO is:

A. An aldehyde

B. A carboxylic acid

C. An alcohol

D. A ketone

Q4 CH_3COCH_3 is:

A. A ketone

B. An ester

C. An aldehyde

D. An ether

10.2 Nomenclature of Aldehydes & Ketones

Q5 In IUPAC names, the suffix used for an aldehyde is:

A. -ol

B. -one

C. -al

D. -oxo

Q6 Which order shows the priority for choosing the suffix (highest first)?

A. Amine > alcohol > ketone > aldehyde

B. Alcohol > aldehyde > ketone > amine

C. Aldehyde > ketone > alcohol > amine

D. Ketone > aldehyde > alcohol > amine

Q7 When a ketone is NOT the highest-priority group, it is named with the prefix:

A. oxo

B. hydroxy

C. keto-al

D. carbonyl

Q8 When numbering the chain of an aldehyde or ketone, the lowest number goes to:

A. The substituents in alphabetical order

B. The C=C double bond

C. The functional group named by the suffix (C=O)

D. The C≡C triple bond

Q9 The root name of the chain is followed by -an, -en or -yn depending on:

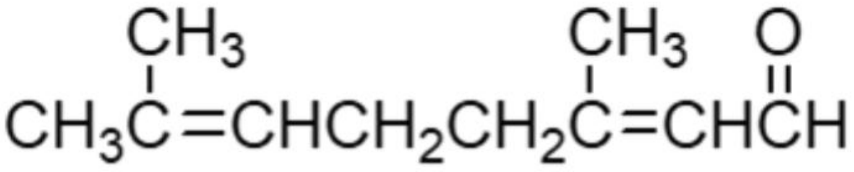
A. Whether the chain has only single bonds, a double bond, or a triple bond

B. Whether the compound is an aldehyde or a ketone

C. The number of substituents

D. The number of oxygen atoms

Q10 Name this compound:



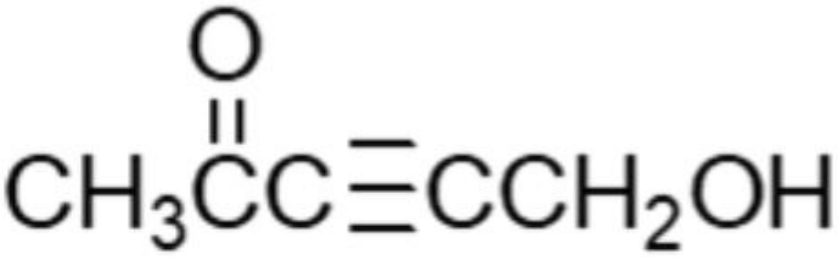
A. 2,6-Dimethyl-3,7-octadienal

B. 2,6-Dimethyl-2,6-octadienal

C. 3,7-Dimethyl-2,6-octadienal

D. 3,7-Dimethyl-2,6-octadienone

Q11 Name this compound:



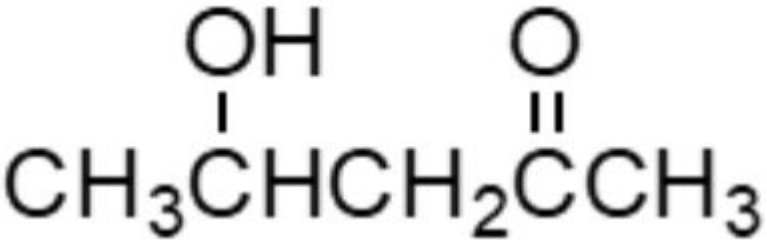
A. 4-Oxo-2-pentyn-1-ol

B. 5-Hydroxy-3-pentyn-2-one

C. 5-Hydroxy-2-pentyn-4-one

D. 1-Hydroxy-2-pentyn-4-one

Q12 Name this compound:



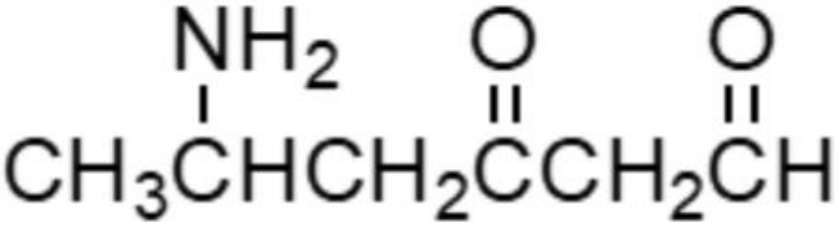
A. 2-Hydroxy-4-pentanone

B. 4-Hydroxy-2-pentanone

C. 2-Hydroxy-4-pentanal

D. 4-Oxo-2-pentanol

Q13 Name this compound:



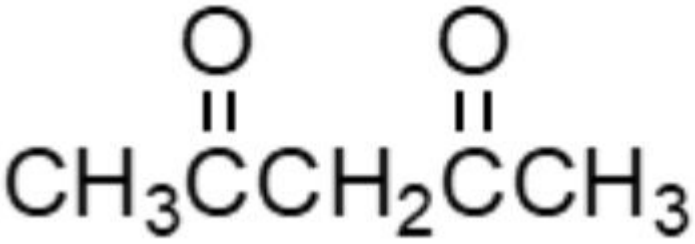
A. 1-Amino-3-oxo-5-hexanal

B. 2-Amino-4-oxohexanal

C. 5-Amino-3-oxohexanal

D. 5-Amino-3-hexanone-1-al

Q14 Name this compound:



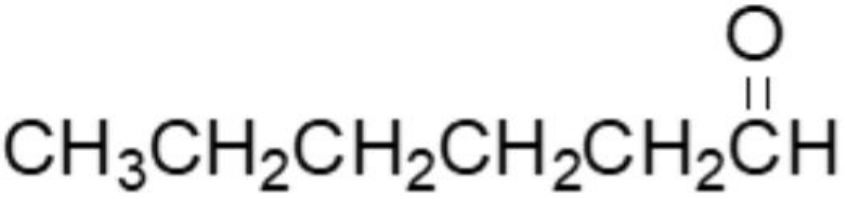
A. 2,4-Pentanedial

B. 1,3-Pentanedione

C. 2,4-Pentanedione

D. 2-Oxo-4-pentanone

Q15 Name this compound:



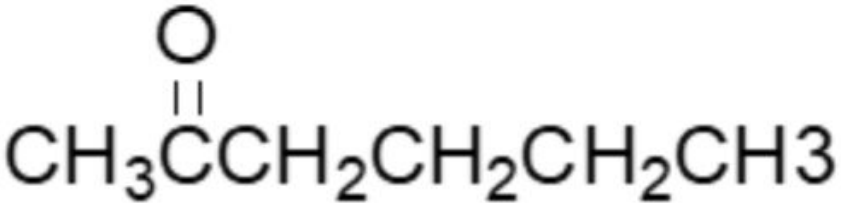
A. 1-Hexanol

B. Pentanal

C. Hexanone

D. Hexanal

Q16 Name this compound:



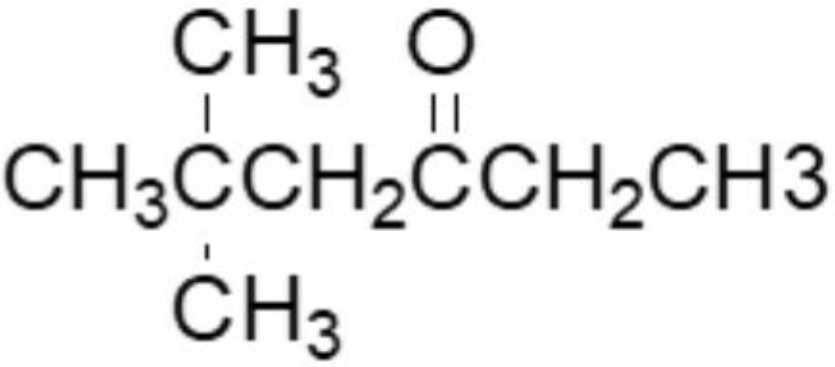
A. 2-Pentanone

B. Hexanal

C. 5-Hexanone

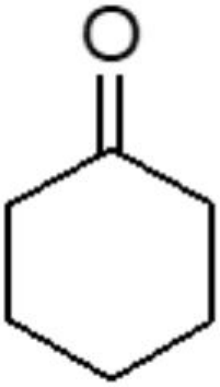
D. 2-Hexanone

Q17 Name this compound:



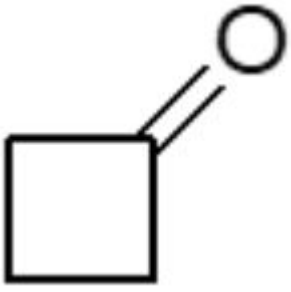
A. 2,2-Dimethyl-4-hexanone
B. 5,5-Dimethyl-3-hexanone
C. 2,2-Dimethyl-3-hexanone
D. 5,5-Dimethyl-4-hexanone

Q18 Name this compound:



A. Cyclohexanol
B. Cyclopentanone
C. Cyclohexanal
D. Cyclohexanone

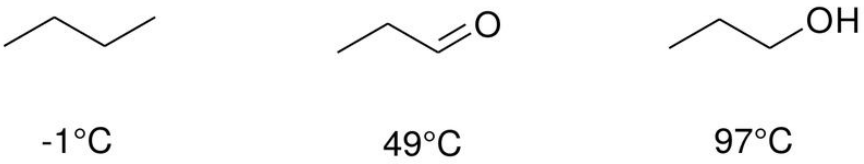
Q19 Name this compound:



A. Cyclobutanol
B. Cyclobutanal
C. Cyclobutanone
D. Cyclopropanone

10.3 Physical Properties

Q20 Butane (−1 °C), propanal (49 °C), 1-propanol (97 °C). The boiling points of aldehydes & ketones are:



A. Higher than alcohols

B. Between alkanes and alcohols of similar molecular weight

C. The same as alkanes

D. Lower than alkanes

Q21 Why do aldehydes & ketones boil higher than alkanes of similar size?

A. The C=O group is polar, so there are dipole–dipole attractions between molecules

B. They are ionic compounds

C. They form hydrogen bonds with each other

D. They have a higher molecular weight

Q22 Why do aldehydes & ketones boil lower than alcohols?

A. They have more carbon atoms

B. They form stronger dipole–dipole forces

C. They are more polar than alcohols

D. They have no O–H, so they cannot hydrogen bond with each other

Q23 Why are small aldehydes & ketones soluble in water?

A. They are H-bond donors

B. The carbonyl O is an H-bond acceptor, so it forms H-bonds with water

C. They are non-polar

D. They react with water to form salts

10.4 Preparation of Aldehydes & Ketones

Q24 Which methods are used to prepare aldehydes & ketones?

A. Hydrogenation of alkynes, dehydration of alcohols, halogenation of alkanes, hydrolysis of ethers

B. Reduction of carboxylic acids only

C. Hydration of alkenes, reduction of alcohols, nitration of benzene, ozonolysis of alkanes

D. Hydration of alkynes, ozonolysis of alkenes, oxidation of alcohols, Friedel–Crafts acylation of benzene

Q25 Hydration of an alkyne needs which reagents?

A. H_2O, H_2SO_4 and $HgSO_4$

B. O_3 then Zn/H_2O

C. $NaBH_4$ then H_3O^+

D. RCOCl and $AlCl_3$

Q26 Hydration of phenylacetylene ($C_6H_5C{\equiv}CH$) with H_2O, H_2SO_4, $HgSO_4$ gives:

A. Styrene

B. Phenylacetaldehyde ($C_6H_5CH_2CHO$)

C. Acetophenone ($C_6H_5COCH_3$), through an enol

D. Benzoic acid

Q27 Ozonolysis of an alkene (O_3, then Zn/H_2O) gives:

A. An alcohol

B. An epoxide

C. An alkyne

D. Two carbonyl compounds (aldehydes and/or ketones) — the C=C is cut

Q28 Oxidation of a 2° alcohol gives:

A. A carboxylic acid

B. A ketone

C. An aldehyde

D. No reaction

Q29 Oxidation of a 3° alcohol gives:

A. A ketone

B. No reaction

C. An aldehyde

D. A carboxylic acid

Q30 A 1° alcohol can be oxidized to:

A. A ketone only

B. An aldehyde or a carboxylic acid, depending on the reagent

C. No reaction

D. An ether

Q31 Friedel–Crafts acylation (benzene + RCOCl, $AlCl_3$) is used to prepare:

A. Aliphatic aldehydes

B. Aromatic ketones

C. Phenols

D. Benzoic acid

10.5 Reaction Sites & Oxidation

Q32 In the carbonyl group, nucleophiles attack:

A. The α-carbon only

B. The C–H bond of the aldehyde

C. The carbonyl oxygen (δ−)

D. The carbonyl carbon (δ+)

Q33 Electrophiles and acids (H^+) attack the carbonyl group at:

A. The carbon

B. The oxygen, which has two non-bonding electron pairs

C. The R group

D. The α-hydrogen

Q34 How can aldehydes and ketones be distinguished in the lab?

A. Aldehydes are easily oxidized under mild conditions, ketones are not (e.g. Tollens' test)

B. Only aldehydes are soluble in water

C. Ketones are easily oxidized, aldehydes are not

D. Only ketones react with $NaBH_4$

Q35 Tollens' reagent is:

A. Copper sulfate in NaOH

B. $KMnO_4$ in water

C. Silver nitrate in ammonium hydroxide, $Ag(NH_3)_2OH$

D. CrO_3 in H_2SO_4

Q36 A positive Tollens' test (silver mirror) means:

A. The ketone is reduced to an alcohol

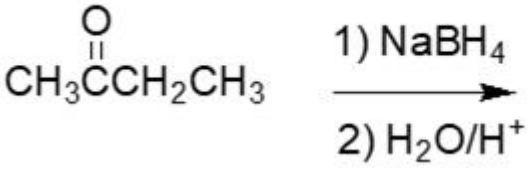
B. Ag is oxidized to Ag^+

C. The aldehyde is reduced to an alcohol

D. The aldehyde is oxidized to a carboxylate and Ag^+ is reduced to metallic silver

10.6 Reduction & Nucleophilic Addition

Q37 Complete the reaction:



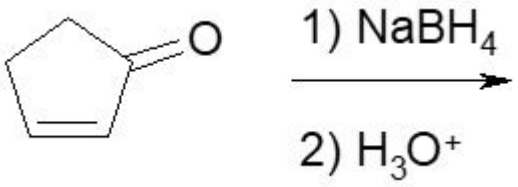
A. 2-Butanol, $CH_3CH(OH)CH_2CH_3$

B. Butane

C. Butanal

D. Butanoic acid

Q38 Complete the reaction:



A. Cyclopentanol — both C=O and C=C are reduced

B. Cyclopentanone — only the C=C is reduced

C. A cyclopentenol — only the C=O is reduced, the C=C stays

D. No reaction

Q39 Base-initiated nucleophilic addition — which step happens first?

A. The nucleophile (Nu^-) attacks the carbonyl carbon

B. The α-H is removed

C. Water leaves

D. H^+ bonds to the carbonyl oxygen

Q40 In the base-initiated mechanism, the intermediate formed after the Nu^- attack is:

A. A trigonal planar enol

B. A free radical

C. A tetrahedral alkoxide ion

D. A carbocation

Q41 Acid-initiated nucleophilic addition — which step happens first?

A. The C=C forms

B. Nu^- attacks the carbonyl carbon

C. The alkoxide takes a proton

D. H^+ bonds to the partially negative carbonyl oxygen

Q42 Addition of an alcohol (R'OH) to an aldehyde first gives a ____, and with a second R'OH and H^+ gives a ____.

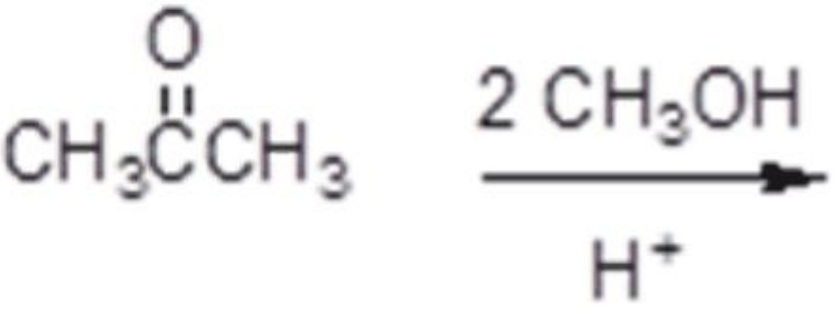
A. Imine; oxime

B. Hydrate; cyanohydrin

C. Acetal; hemiacetal

D. Hemiacetal; acetal

Q43 Complete the reaction:



A. Methyl acetate

B. Isopropanol

C. 2-Methoxy-2-propanol only

D. 2,2-Dimethoxypropane, $(CH_3)_2C(OCH_3)_2$ — an acetal

Q44 Addition of water to an aldehyde or ketone gives a:

A. Hydrate (gem-diol), $R_2C(OH)_2$

B. Enol

C. Carboxylic acid

D. Hemiacetal

Q45 Addition of a Grignard reagent (R'MgX) to a ketone, then H_3O^+, gives:

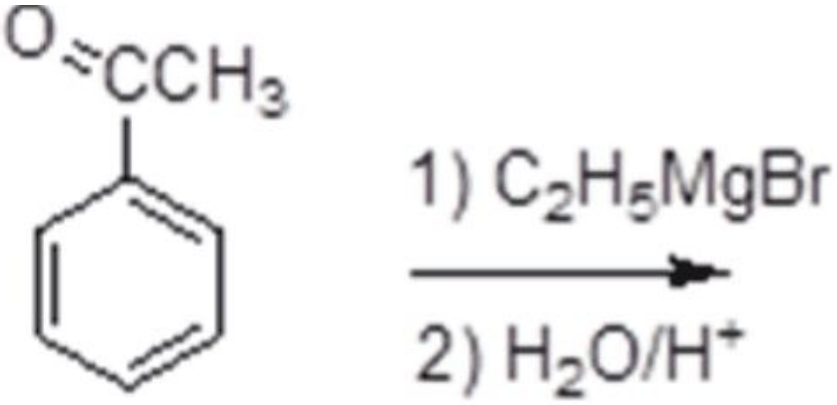
A. A 3° alcohol

B. An aldehyde

C. A 2° alcohol

D. A 1° alcohol

Q46 Complete the reaction:



A. 1-Phenyl-1-propanol

B. 2-Phenylbutane

C. Propiophenone

D. 2-Phenyl-2-butanol, $C_6H_5C(OH)(CH_3)CH_2CH_3$

Q47 Addition of HCN to an aldehyde or ketone gives a:

A. Nitrile, RCN

B. Cyanohydrin, $R_2C(OH)CN$

C. Imine

D. Amide

10.7 Nitrogen Derivatives

Q48 An aldehyde or ketone + a 1° amine (RNH_2) gives:

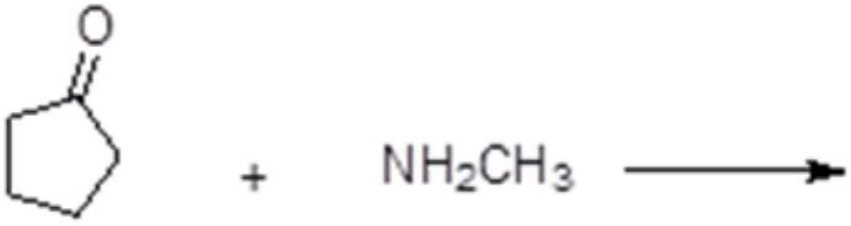
A. An amide

B. A cyanohydrin

C. A 2° amine

D. An imine (Schiff base), $R_2C{=}NR' + H_2O$

Q49 Complete the reaction:



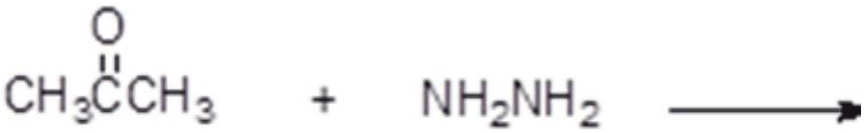
A. Cyclopentanol

B. N-Methylcyclopentylamine

C. N-Methylcyclopentanimine ($C{=}N{-}CH_3$) + H_2O

D. Cyclopentanone oxime

Q50 Complete the reaction:



A. Acetone oxime, $(CH_3)_2C{=}NOH$

B. Acetone hydrazone, $(CH_3)_2C{=}N{-}NH_2 + H_2O$

C. Isopropanol

D. 2-Propanamine

Q51 Which pair is correct?

A. Phenylhydrazine → imine

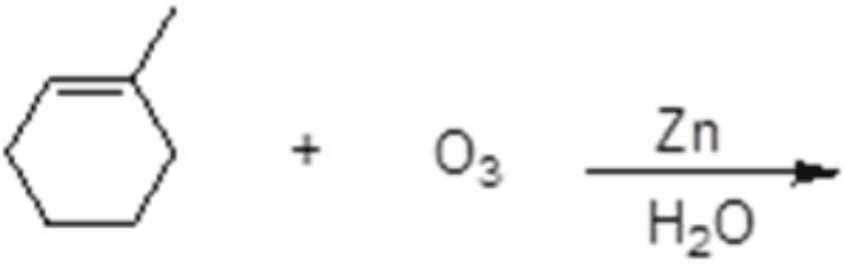
B. Hydroxylamine (NH_2OH) → oxime (C=N–OH)

C. Hydrazine (NH_2NH_2) → oxime

D. 1° amine → hydrazone

10.8 Ozonolysis Practice

Q52 Complete the reaction:



A. 1-Methylcyclohexanol

B. Two molecules: cyclohexanone + formaldehyde

C. Heptanoic acid

D. 6-Oxoheptanal, $CH_3CO(CH_2)_4CHO$ — one molecule

10.9 Tautomerism & α-Hydrogens

Q53 Aldehydes & ketones can exist as a mixture of two forms called:

A. Keto and enol forms

B. Chair and boat forms

C. Cis and trans forms

D. R and S forms

Q54 What is the difference between the keto and enol forms?

A. The molecular formula

B. The position of a hydrogen and of the double bond

C. The number of carbon atoms

D. The arrangement in space only (stereoisomers)

Q55 Tautomers are:

A. Resonance structures of the same molecule

B. Structural isomers that differ in the position of a hydrogen (and a double bond) and rapidly interconvert

C. Stereoisomers that are mirror images

D. Molecules with different molecular formulas

Q56 An α-hydrogen is:

A. A hydrogen on the carbon next to the carbonyl group

B. The hydrogen of an O–H group

C. Any hydrogen on a benzene ring

D. The hydrogen on the carbonyl carbon of an aldehyde

Q57 Why are α-hydrogens acidic?

A. The anion formed (enolate) is stabilized by resonance — the negative charge is shared with the oxygen

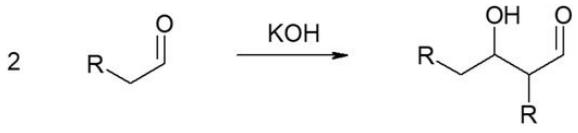
B. Because they are bonded to oxygen

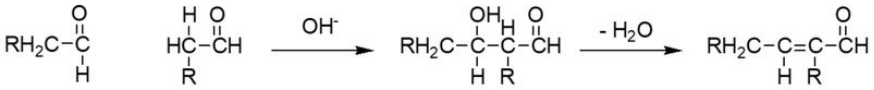
C. Because they are part of an O–H group

D. Because the α-carbon is sp hybridized

10.10 Aldol Condensation

Q58 What are the three steps of the aldol reaction (base, KOH)?





A. 1) Oxidation · 2) reduction · 3) hydration

B. 1) Nu^- attacks · 2) H_2O is added · 3) the C=C is cut

C. 1) OH^- removes an α-H → enolate · 2) enolate attacks the C=O of a second molecule → alkoxide · 3) alkoxide takes H^+ from water → β-hydroxy aldehyde

D. 1) H^+ adds to O · 2) water leaves · 3) carbocation forms

Q59 The product of the aldol reaction is a:

A. Diol

B. Ester

C. α-Hydroxy acid

D. β-Hydroxy aldehyde (or ketone)

Q60 If the aldol product loses water ($-H_2O$), the product is:

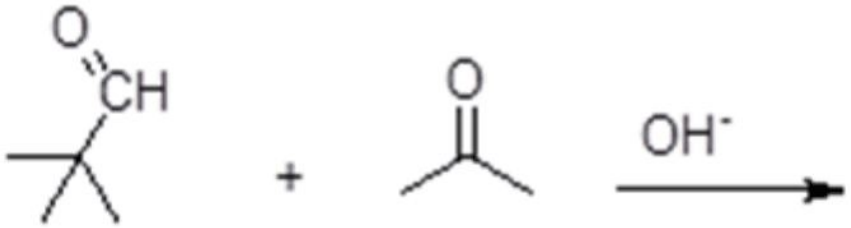
A. A carboxylic acid

B. A saturated alcohol

C. An α,β-unsaturated aldehyde (C=C next to the C=O)

D. An alkyne

Q61 Complete the reaction (crossed aldol):



A. $(CH_3)_3C–CH(OH)–CH_2–CO–CH_3$ (4-hydroxy-5,5-dimethyl-2-hexanone)

B. No reaction — neither compound has α-H

C. $(CH_3)_3C–CO–CH_3$

D. $(CH_3)_3C–COOH$

Answers · Lecture 10

10.1 Structure of Aldehydes & Ketones

Q1 **A. Aldehydes have at least one H on the carbonyl carbon; ketones have two carbon groups on the carbonyl carbon**

General structure R1–CO–R2: aldehyde → R1 or R2 = H; ketone → R1 and R2 are both carbon groups.

Q2 **C. The carbonyl group (C=O)**

Aldehydes and ketones are both carbonyl compounds.

Q3 **A. An aldehyde**

–CHO = H on the carbonyl carbon → aldehyde (propanal).

Q4 **A. A ketone**

The carbonyl carbon has two CH_3 groups → ketone (propanone, acetone).

10.2 Nomenclature of Aldehydes & Ketones

Q5 **C. -al**

Aldehyde → suffix **-al** (prefix oxo). Ketone → **-one** (prefix oxo). Alcohol → -ol (prefix hydroxy). Amine → -amine (prefix amino).

Q6 **C. Aldehyde > ketone > alcohol > amine**

Aldehyde > ketone > alcohol > amine. The highest one gets the suffix; the others are named as prefixes.

Q7 **A. oxo**

Both aldehyde and ketone use the prefix **oxo** when they are not the main group.

Q8 **C. The functional group named by the suffix (C=O)**

Number the chain to give the lowest number to the suffix group. Next come C–C multiple bonds (C=C before C≡C when there is a tie), then the prefixes (alphabetical order).

Q9 **A. Whether the chain has only single bonds, a double bond, or a triple bond**

-an = all C–C single bonds, -en = double bond, -yn = triple bond.

Q10 **C. 3,7-Dimethyl-2,6-octadienal**

Longest chain = 8 C (**oct**) · two C=C (**octadien**) · aldehyde (**octadienal**) · number from the CHO: C=C at 2 and 6 · methyls at 3 and 7 → **3,7-dimethyl-2,6-octadienal**.

Q11 **B. 5-Hydroxy-3-pentyn-2-one**

Chain = 5 C (**pent**) · triple bond (**pentyn**) · ketone is the main group (**pentynone**) · number from the ketone end: C=O at 2, C≡C at 3 · OH at 5 → **5-hydroxy-3-pentyn-2-one**.

Q12 **B. 4-Hydroxy-2-pentanone**

Ketone has priority over alcohol → suffix -one, OH → hydroxy. Number from the end nearest the C=O: C=O at 2, OH at 4 → **4-hydroxy-2-pentanone**.

Q13 **C. 5-Amino-3-oxohexanal**

Aldehyde is the highest priority → suffix -al, CHO = C1. Ketone at C3 → **oxo**, NH_2 at C5 → **amino**. Prefixes in alphabetical order → **5-amino-3-oxohexanal**.

Q14 **C. 2,4-Pentanedione**

5-carbon chain with two ketone groups at C2 and C4 → **2,4-pentanedione**.

Q15 **D. Hexanal**

6 carbons including the CHO carbon → **hexanal**.

Q16 **D. 2-Hexanone**

6-carbon chain, C=O at C2 (lowest number) → **2-hexanone**.

Q17 **B. 5,5-Dimethyl-3-hexanone**

Longest chain = 6 C. Number from the right so the C=O gets the lowest number (3, not 4) → two methyls on C5 → **5,5-dimethyl-3-hexanone**.

Q18 **D. Cyclohexanone**

6-membered ring with a ketone → **cyclohexanone**.

Q19 **C. Cyclobutanone**

4-membered ring with a ketone → **cyclobutanone**.

10.3 Physical Properties

Q20 **B. Between alkanes and alcohols of similar molecular weight**

Intermediate between alkanes and alcohols.

Q21 **A. The C=O group is polar, so there are dipole–dipole attractions between molecules**

The polar C=O gives dipole–dipole attractions; alkanes are non-polar.

Q22 **D. They have no O–H, so they cannot hydrogen bond with each other**

Alcohols have O–H and form hydrogen bonds between molecules; aldehydes & ketones cannot (no H on O).

Q23 **B. The carbonyl O is an H-bond acceptor, so it forms H-bonds with water**

Smaller compounds are soluble in water because the C=O oxygen accepts H-bonds from water.

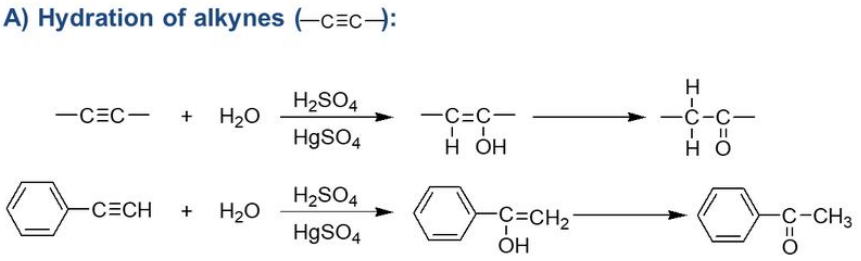
10.4 Preparation of Aldehydes & Ketones

Q24 **D. Hydration of alkynes, ozonolysis of alkenes, oxidation of alcohols, Friedel–Crafts acylation of benzene**

The 4 methods: (A) hydration of alkynes, (B) ozonolysis of alkenes, (C) oxidation of alcohols, (D) acylation of benzene (Friedel–Crafts).

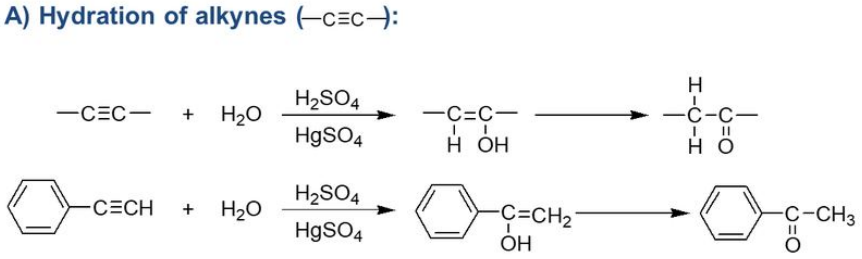
Q25 **A. H_2O, H_2SO_4 and $HgSO_4$**

Alkyne + H_2O with H_2SO_4/$HgSO_4$ → enol → ketone (aldehyde only from ethyne).



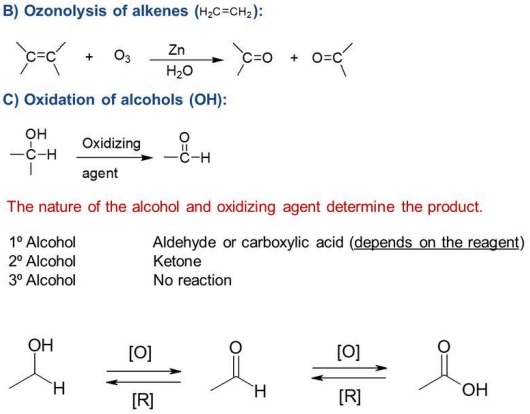
Q26 **C. Acetophenone ($C_6H_5COCH_3$), through an enol**

Markovnikov addition: OH goes on the carbon attached to the ring → enol → tautomerizes to the ketone $C_6H_5COCH_3$.



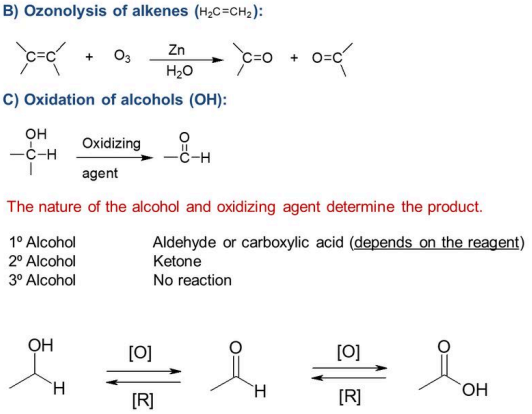
Q27 **D. Two carbonyl compounds (aldehydes and/or ketones) — the C=C is cut**

The C=C breaks and each carbon becomes C=O.



Q28 **B. A ketone**

1° alcohol → aldehyde or carboxylic acid (depends on the reagent) · 2° alcohol → **ketone** · 3° alcohol → no reaction.



Q29 **B. No reaction**

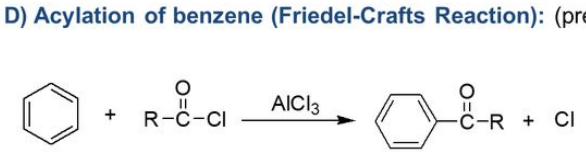
3° alcohols have no H on the C–OH carbon, so they are not oxidized.

Q30 **B. An aldehyde or a carboxylic acid, depending on the reagent**

Mild oxidizing agent → aldehyde; strong oxidizing agent → oxidized further to the carboxylic acid.

Q31 **B. Aromatic ketones**

Benzene + acyl chloride with $AlCl_3$ → aryl ketone (C_6H_5COR) + HCl.



10.5 Reaction Sites & Oxidation

Q32 **D. The carbonyl carbon (δ+)**

O is more electronegative → polar C=O. Nucleophiles and bases are attracted to the C (δ+); electrophiles and acids are attracted to the O (δ−), which has two non-bonding electron pairs.

Q33 **B. The oxygen, which has two non-bonding electron pairs**

The O is δ− and has 2 lone pairs, so E^+ and H^+ go to the O.

Q34 **A. Aldehydes are easily oxidized under mild conditions, ketones are not (e.g. Tollens' test)**

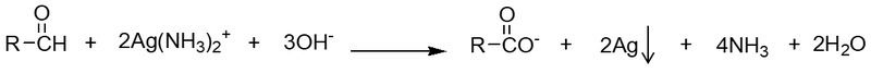
They differ in their susceptibility to oxidation: aldehydes are oxidized by mild oxidizing agents; ketones do not react.

Q35 **C. Silver nitrate in ammonium hydroxide, $Ag(NH_3)_2OH$**

Tollens' reagent = $AgNO_3$ in NH_4OH → $[Ag(NH_3)_2]OH$.

Q36 **D. The aldehyde is oxidized to a carboxylate and Ag^+ is reduced to metallic silver**

$R–CHO + 2Ag(NH_3)_2^+ + 3OH^- \rightarrow R–COO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$. The silver deposits on the clean glass of the test tube.



10.6 Reduction & Nucleophilic Addition

Q37 **A. 2-Butanol, $CH_3CH(OH)CH_2CH_3$**

$NaBH_4$ reduces the ketone C=O to a 2° alcohol → **2-butanol**.

Q38 **C. A cyclopentenol — only the C=O is reduced, the C=C stays**

$NaBH_4$ reduces the C=O to C–OH but does not reduce the C=C.

Q39 **A. The nucleophile (Nu^-) attacks the carbonyl carbon**

1) Nu^- attacks the carbonyl C and gives both electrons for the new Nu–C bond. 2) The π electrons move to the O → alkoxide anion (tetrahedral). 3) The O^- takes H^+ from H–Nu → alcohol product.

Q40 **C. A tetrahedral alkoxide ion**

The carbonyl C goes from trigonal planar to tetrahedral; the O carries the negative charge (alkoxide).

Q41 **D. H^+ bonds to the partially negative carbonyl oxygen**

1) H^+ bonds to the δ− O → the carbon becomes a carbocation (C^+). 2) The C^+ attracts the Nu to the carbon.

Q42 **D. Hemiacetal; acetal**

RCHO + R'OH ⇌ hemiacetal RCH(OH)OR'. Hemiacetal + R'OH (H^+) → acetal $RCH(OR')_2 + H_2O$.

Q43 **D. 2,2-Dimethoxypropane, $(CH_3)_2C(OCH_3)_2$ — an acetal**

Ketone + 2 CH_3OH (H^+) → acetal $(CH_3)_2C(OCH_3)_2 + H_2O$.

Q44 **A. Hydrate (gem-diol), $R_2C(OH)_2$**

$C=O + H_2O \rightleftharpoons R_2C(OH)_2$ — two OH groups on the same carbon.

Q45 **A. A 3° alcohol**

Formaldehyde → 1° alcohol · other aldehydes → 2° alcohol · ketones → 3° alcohol.

Q46 **D. 2-Phenyl-2-butanol, $C_6H_5C(OH)(CH_3)CH_2CH_3$**

Acetophenone (ketone) + C_2H_5MgBr → the ethyl adds to the carbonyl C; H_3O^+ gives the 3° alcohol **2-phenyl-2-butanol**.

Q47 **B. Cyanohydrin, $R_2C(OH)CN$**

CN^- attacks the C, the O takes H^+ → OH and CN on the same carbon (cyanohydrin).

10.7 Nitrogen Derivatives

Q48 **D. An imine (Schiff base), $R_2C{=}NR' + H_2O$**

Addition of RNH_2 then loss of water → C=N (imine).

Q49 **C. N-Methylcyclopentanimine ($C{=}N{-}CH_3$) + H_2O**

Cyclopentanone + CH_3NH_2 → imine: the O is replaced by =N–CH_3, and H_2O is lost.

Q50 **B. Acetone hydrazone, $(CH_3)_2C{=}N{-}NH_2 + H_2O$**

Ketone + hydrazine (NH_2NH_2) → hydrazone (C=N–NH_2) + H_2O.

Q51 **B. Hydroxylamine (NH_2OH) → oxime (C=N–OH)**

1° amine → imine (C=NR) · hydroxylamine → oxime (C=NOH) · hydrazine → hydrazone ($C{=}NNH_2$) · phenylhydrazine → phenylhydrazone ($C{=}NNHC_6H_5$).

10.8 Ozonolysis Practice

Q52 **D. 6-Oxoheptanal, $CH_3CO(CH_2)_4CHO$ — one molecule**

The C=C is inside the ring, so cutting it opens the ring into one chain: the methyl carbon becomes a ketone and the other carbon becomes an aldehyde → **6-oxoheptanal**.

10.9 Tautomerism & α-Hydrogens

Q53 **A. Keto and enol forms**

Keto form: C=O with an H on the α-carbon. Enol form: C=C–OH (ene + ol). The keto form is usually the major one.

Q54 **B. The position of a hydrogen and of the double bond**

The H moves from the α-carbon to the O, and the double bond moves from C=O to C=C.

Q55 **B. Structural isomers that differ in the position of a hydrogen (and a double bond) and rapidly interconvert**

Tautomers are real, different compounds in equilibrium — not resonance forms.

Q56 **A. A hydrogen on the carbon next to the carbonyl group**

The carbon next to C=O is the α-carbon; its H atoms are α-hydrogens.

Q57 **A. The anion formed (enolate) is stabilized by resonance — the negative charge is shared with the oxygen**

Removing an α-H gives an enolate ion; the negative charge is delocalized onto the electronegative O, so α-H are much more acidic than normal C–H.

10.10 Aldol Condensation

Q58 **C. 1) OH^- removes an α-H → enolate · 2) enolate attacks the C=O of a second molecule → alkoxide · 3) alkoxide takes H^+ from water → β-hydroxy aldehyde**

Enolate formation → nucleophilic attack on the second carbonyl → protonation. The product is an aldol (ald + ol): a β-hydroxy aldehyde. With heat it loses H_2O → α,β-unsaturated aldehyde.

Q59 **D. β-Hydroxy aldehyde (or ketone)**

"Aldol" = aldehyde + alcohol in the same molecule; the OH is on the β-carbon.

Q60 **C. An α,β-unsaturated aldehyde (C=C next to the C=O)**

Dehydration forms a C=C between the α and β carbons → condensation product.

Q61 **A. $(CH_3)_3C–CH(OH)–CH_2–CO–CH_3$ (4-hydroxy-5,5-dimethyl-2-hexanone)**

2,2-Dimethylpropanal has **no α-H**, so only acetone forms the enolate. The acetone enolate attacks the aldehyde C=O → β-hydroxy ketone. (With heat it can lose H_2O → $(CH_3)_3C–CH=CH–CO–CH_3$.)

11 Carboxylic Acids & Derivatives

In this lecture

1. Structure & Nomenclature of Carboxylic Acids 10 questions
2. Physical Properties 4 questions
3. Acidity of Carboxylic Acids 9 questions
4. Preparation of Carboxylic Acids 4 questions
5. Carboxylic Acid Derivatives: Structure & Names 5 questions
6. Preparation of Acid Derivatives 4 questions
7. Nucleophilic Acyl Substitution 13 questions
8. Reactions with Grignard Reagents & $LiAlH_4$ 5 questions

11.1 Structure & Nomenclature of Carboxylic Acids

Q1 What is the general structure of a carboxylic acid?

A. R–CHO — a carbonyl with an H

B. R–COOH — a carbonyl (C=O) and a hydroxyl (–OH) on the same carbon

C. R–O–R — oxygen between two carbons

D. R–CO–R — a carbonyl between two carbons

Q2 In IUPAC names of carboxylic acids, the -e of the alkane is replaced by:

A. -oic acid

B. -ol

C. -al

D. -one

Q3 When numbering the chain of a carboxylic acid, the COOH carbon is:

A. Given the highest number

B. Not counted in the chain

C. C2 (the α-carbon)

D. Always C1

Q4 In common names, the carbon next to the COOH is called:

A. γ (gamma)

B. ω (omega)

C. α (alpha)

D. β (beta)

Q5 In a molecule with a COOH and a ketone, the ketone is named with the prefix:

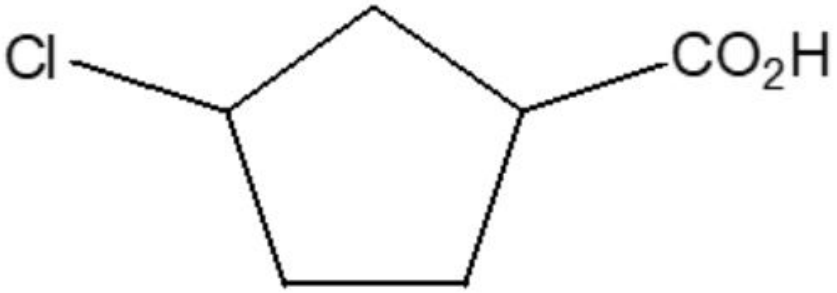
A. hydroxy

B. keto-one

C. carbonyl

D. oxo

Q6 Name this compound:



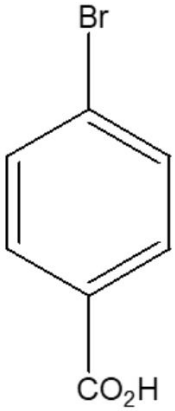
A. 4-Chlorocyclopentanecarboxylic acid

B. 1-Chloro-3-cyclopentanoic acid

C. 3-Chlorocyclopentanecarboxylic acid

D. 3-Chlorocyclopentanoic acid

Q7 Name this compound:



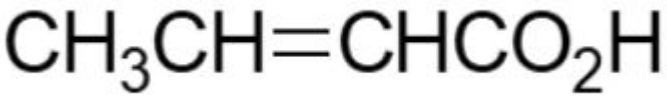
A. 3-Bromobenzoic acid (m-bromobenzoic acid)

B. 4-Bromophenol

C. 4-Bromobenzoic acid (p-bromobenzoic acid)

D. 2-Bromobenzoic acid (o-bromobenzoic acid)

Q8 Name this compound:



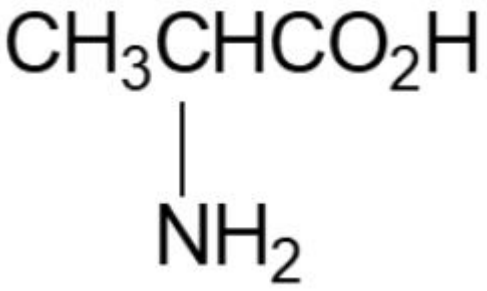
A. 2-Butenal

B. 2-Butenoic acid

C. Butanoic acid

D. 3-Butenoic acid

Q9 Name this compound:



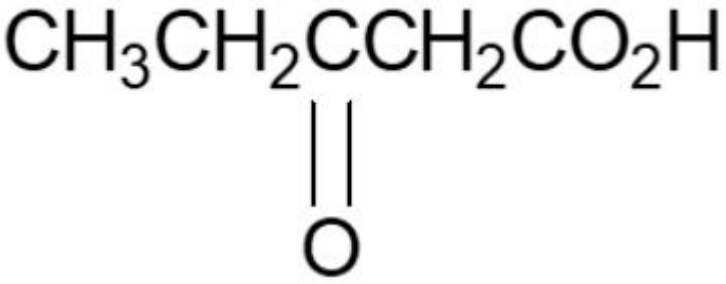
A. 2-Carboxypropanamine

B. 1-Aminopropanoic acid

C. 2-Aminopropanoic acid

D. 3-Aminopropanoic acid

Q10 Name this compound:



A. 3-Oxopentanoic acid

B. 3-Oxopentanal

C. 4-Oxopentanoic acid

D. 3-Pentanone-1-oic acid

11.2 Physical Properties

Q11 How do the boiling points of carboxylic acids compare with alcohols of similar molecular weight?

A. Carboxylic acids do not have boiling points

B. They are the same

C. Carboxylic acids boil higher

D. Carboxylic acids boil lower

Q12 Carboxylic acids have high boiling points mainly because:

A. They form hydrogen-bonded dimers (two H-bonds between two molecules)

B. They are non-polar

C. They cannot form hydrogen bonds

D. They are ionic in the pure liquid

Q13 What about the solubility of carboxylic acids in water?

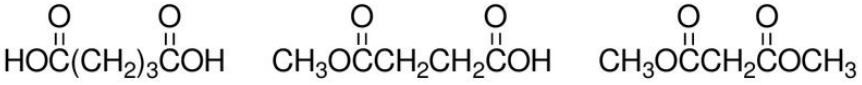
A. They dissolve only because they are ionic

B. All carboxylic acids are insoluble in water

C. Solubility increases as the chain gets longer

D. Small acids (about 1–4 carbons) are water-soluble because of H-bonding; solubility decreases as the carbon chain gets longer

Q14 These three compounds have the same molecular formula ($C_5H_8O_4$). Arrange them in order of increasing boiling point.



A. $CH_3OCOCH_2COOCH_3 < CH_3OCOCH_2CH_2COOH < HOOC(CH_2)_3COOH$

B. $CH_3OCOCH_2CH_2COOH < CH_3OCOCH_2COOCH_3 < HOOC(CH_2)_3COOH$

C. $HOOC(CH_2)_3COOH < CH_3OCOCH_2CH_2COOH < CH_3OCOCH_2COOCH_3$

D. All three have the same boiling point

11.3 Acidity of Carboxylic Acids

Q15 Order of acidity (most → least):

A. Phenol > carboxylic acid > alcohol

B. Alcohol > phenol > carboxylic acid

C. Carboxylic acid > phenol > alcohol

D. Carboxylic acid > alcohol > phenol

Q16 Why are carboxylic acids acidic?

A. Because the alkyl group donates electrons

B. Because they contain a C=C bond

C. The carboxylate ion ($RCOO^-$) is stabilized by resonance — the negative charge is shared equally by two oxygens

D. Because the O–H bond is very strong

Q17 Which reagent reacts with a carboxylic acid but NOT with a phenol?

A. NaOH

B. HCl

C. $NaHCO_3$ (sodium bicarbonate, weak base) — the acid gives CO_2 bubbles

D. H_2O

Q18 Which statement about NaOH is correct?

A. NaOH (strong base) reacts with both carboxylic acids and phenols to form water-soluble salts

B. NaOH does not react with any of them

C. NaOH reacts with carboxylic acids only

D. NaOH reacts only with alcohols

Q19 What can you conclude from these pKa values?

	HCO_2H	CH_3CO_2H	$O_2N\text{-}CH_2CO_2H$
Pka	3.75	4.76	1.68

	FCH_2CO_2H	$ClCH_2CO_2H$	$BrCH_2CO_2H$	ICH_2CO_2H	CH_3CO_2H
Pka	2.59	2.87	2.90	3.17	4.75

	$CH_3CH_2\underset{Cl}{C}HCO_2H$	$CH_3\underset{Cl}{C}HCH_2CO_2H$	$\underset{Cl}{C}H_2CH_2CH_2CO_2H$	$CH_3CH_2CH_2CO_2H$
Pka	2.86	4.05	4.52	4.82

A. The position of the substituent has no effect

B. Alkyl groups increase acidity

C. Electron-withdrawing groups decrease acidity

D. Electron-withdrawing groups increase acidity; the closer they are to the COOH (and the more there are), the stronger the acid

Q20 Which is the strongest acid?

A. $BrCH_2COOH$

B. FCH_2COOH

C. ICH_2COOH

D. $ClCH_2COOH$

Q21 Why is formic acid (HCOOH, pKa 3.75) stronger than acetic acid (CH_3COOH, pKa 4.76)?

A. Formic acid has more oxygen atoms

B. The CH_3 group donates electrons and destabilizes the carboxylate ion

C. Acetic acid forms more hydrogen bonds

D. The CH_3 group withdraws electrons

Q22 Arrange from least acidic to most acidic:

$CH_3\underset{F}{C}HCO_2H$ $\quad$ $CH_2\underset{F}{\underset{|}{C}}HCO_2H$ $\quad$ $CH_3\underset{Cl}{C}HCO_2H$ $\quad$ $\underset{Cl}{C}H_2CH_2CO_2H$

A. $CH_2FCHFCOOH < CH_3CHFCOOH < CH_3CHClCOOH < ClCH_2CH_2COOH$

B. $ClCH_2CH_2COOH < CH_3CHClCOOH < CH_3CHFCOOH < CH_2FCHFCOOH$

C. $CH_3CHFCOOH < CH_3CHClCOOH < ClCH_2CH_2COOH < CH_2FCHFCOOH$

D. $CH_3CHClCOOH < ClCH_2CH_2COOH < CH_3CHFCOOH < CH_2FCHFCOOH$

Q23 Arrange from least acidic to most acidic:

$CH_3CH_2CH_2\overset{\overset{O}{\|}}{C}CO_2H$ $\quad$ $CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CO_2H$ $\quad$ $CH_3\overset{\overset{O}{\|}}{C}CH_2CH_2CO_2H$

A. 3-Oxopentanoic acid < 2-oxopentanoic acid < 4-oxopentanoic acid

B. 4-Oxopentanoic acid < 3-oxopentanoic acid < 2-oxopentanoic acid

C. They are all equally acidic

D. 2-Oxopentanoic acid < 3-oxopentanoic acid < 4-oxopentanoic acid

11.4 Preparation of Carboxylic Acids

Q24 Which methods are used to prepare carboxylic acids?

A. Oxidation of alkylbenzenes, oxidation of 1° alcohols, hydrolysis of nitriles

B. Dehydration of alcohols only

C. Reduction of esters, hydration of alkenes, ozonolysis of alkynes

D. Oxidation of 3° alcohols, hydrogenation of ketones, nitration of benzene

Q25 Oxidation of toluene ($C_6H_5CH_3$) with $KMnO_4$ (heat) gives:

A. Benzoic acid (C_6H_5COOH)

B. Benzyl alcohol

C. Benzaldehyde

D. Phenol

Q26 Oxidation of a 1° alcohol with a strong oxidizing agent (e.g. CrO_3/H^+ or $KMnO_4$) gives:

A. A ketone

B. An ether

C. No reaction

D. A carboxylic acid

Q27 Hydrolysis of a nitrile (R–C≡N) with H_3O^+ and heat gives:

A. R–COOH + NH_4^+

B. R–OH + HCN

C. R–CHO + NH_3

D. R–CH_2NH_2

11.5 Carboxylic Acid Derivatives: Structure & Names

Q28 Which are the carboxylic acid derivatives?

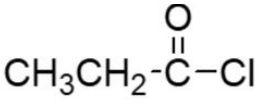
A. Alkenes, alkynes, epoxides, acetals

B. Aldehydes, ketones, alcohols, ethers

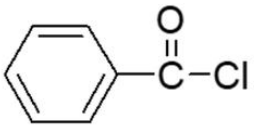
C. Alkyl halides, amines, nitriles, phenols

D. Acid chlorides (RCOCl), acid anhydrides (RCO–O–COR), esters (RCOOR'), amides ($RCONH_2$)

Q29 How do you name an acid chloride?



Propanoyl chloride



Benzoyl chloride

A. Change -oic acid to -amide

B. Change -oic acid to -oate

C. Add the word chloride after the acid name

D. Change -ic acid to -yl chloride (propanoic acid → propanoyl chloride)

Q30 How do you name an acid anhydride (from propanoic or benzoic acid)?

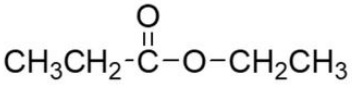
A. Propanoyl anhydride, benzoyl anhydride

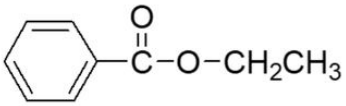
B. Propanoate anhydride

C. Dipropanoic acid

D. Replace the word acid with anhydride: propanoic anhydride, benzoic anhydride

Q31 Name these esters:





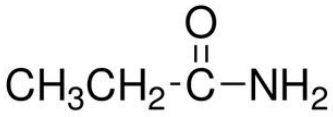
A. Ethyl propanoate and ethyl benzoate

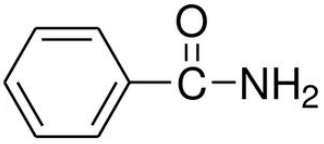
B. Ethyl propanoic acid and ethyl benzoic acid

C. Propanoyl ethane and benzoyl ethane

D. Propyl ethanoate and benzyl ethanoate

Q32 Name these amides:





A. Aminopropanoic acid and aminobenzoic acid

B. Propanoyl amine and benzoyl amine

C. Propanamide and benzamide

D. Propanamine and benzamine

11.6 Preparation of Acid Derivatives

Q33 Acid chlorides are prepared from carboxylic acids with:

A. H_2O

B. $SOCl_2$ (thionyl chloride) or PCl_5

C. $NaBH_4$

D. NaOH

Q34 An acid anhydride can be prepared from:

A. An amide + $LiAlH_4$

B. An alcohol + $KMnO_4$

C. An acid chloride + a carboxylate salt ($RCOCl + RCOO^-Na^+$)

D. An ester + water

Q35 Fischer esterification is:

A. $RCOCl + NH_3 \rightarrow RCONH_2$

B. $RCOOR' + H_2O \rightarrow RCOOH + R'OH$ only in base

C. $RCOOH + R'OH$ with an acid catalyst (H^+) $\rightleftharpoons RCOOR' + H_2O$

D. $RCOOH + NaOH \rightarrow RCOO^-Na^+ + H_2O$

Q36 Amides are best prepared from:

A. An alkene + NH_3

B. An acid chloride + ammonia or an amine

C. An ester + $LiAlH_4$

D. An alcohol + NH_3

11.7 Nucleophilic Acyl Substitution

Q37 The general equation of nucleophilic acyl substitution is:

A. R–CO–L + Nu^- → R–C(OH)(Nu)–L

B. R–CO–L + Nu^- → R–CO–Nu + L^-

C. R–CO–L → R–L + CO

D. R–CO–L + H_2 → R–CH_2–L

Q38 Order of reactivity of the acid derivatives (highest → lowest):

A. Amide > ester > anhydride > acid chloride

B. Acid anhydride > acid chloride > amide > ester

C. Ester > acid chloride > amide > anhydride

D. Acid chloride > acid anhydride > ester > amide

Q39 Acid chloride + H_2O gives:

A. Amide + HCl

B. No reaction

C. Ester + HCl

D. Carboxylic acid + HCl

Q40 Acid chloride + R'OH gives:

A. Amide + HCl

B. Anhydride + HCl

C. Ester + HCl

D. Carboxylic acid + R'Cl

Q41 Acid chloride + R'NH_2 gives:

A. Amine + CO_2

B. Amide (RCONHR') + HCl

C. Ester + HCl

D. Carboxylic acid + R'NH_3^+

Q42 Acid anhydride + H_2O gives:

A. No reaction

B. Amide + acid

C. Ester + acid

D. Two carboxylic acids

Q43 Acid anhydride + R'OH gives:

A. Ester + carboxylic acid

B. Two esters

C. No reaction

D. Ester + HCl

Q44 Acid anhydride + $R'NH_2$ gives:

A. No reaction

B. Ester + amine

C. Amide + carboxylic acid

D. Two amides

Q45 Ester + H_2O (acid or base) gives:

A. Carboxylic acid (or carboxylate in base) + alcohol

B. Two alcohols

C. Amide + alcohol

D. Ketone + alcohol

Q46 Ester + another alcohol (R''OH) gives:

A. No reaction

B. A carboxylic acid + an ether

C. An amide

D. A new ester + R'OH (transesterification)

Q47 Ester + $R''NH_2$ gives:

A. Amide + alcohol (R'OH)

B. Carboxylic acid + amine

C. Ester + amine

D. No reaction

Q48 Amide + H_2O (acid or base, heat) gives:

A. No reaction under any conditions

B. Carboxylic acid (or its salt) + amine / NH_3

C. Ester + NH_3

D. Acid chloride

Q49 Amide + R'OH or $R'NH_2$ gives:

A. No reaction — amides are the least reactive derivative

B. A new amide

C. An ester

D. A carboxylic acid

11.8 Reactions with Grignard Reagents & $LiAlH_4$

Q50 Ester + 2 R"MgX, then H_3O^+, gives:

A. A 1° alcohol

B. A 3° alcohol with two R" groups + R'OH

C. A ketone only

D. A carboxylic acid

Q51 Ester + $LiAlH_4$, then H_3O^+, gives:

A. An aldehyde only

B. A 1° alcohol (RCH_2OH) + R'OH

C. An amine

D. A 2° alcohol

Q52 Amide ($RCONR_2$) + $LiAlH_4$, then H_2O/H^+, gives:



A. A 1° alcohol

B. An amine, RCH_2NR_2

C. A carboxylic acid

D. An aldehyde

Q53 Complete the reaction:



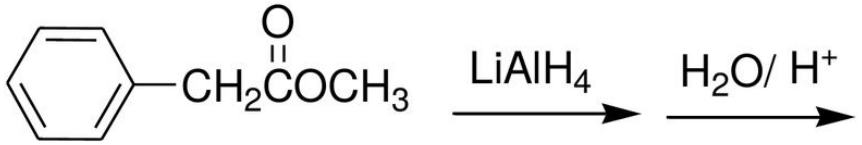
A. Benzoic acid + ethane

B. Acetophenone, $C_6H_5COCH_3$ + CH_3CH_2OH

C. 1-Phenylethanol

D. 2-Phenyl-2-propanol, $C_6H_5C(OH)(CH_3)_2$ + CH_3CH_2OH

Q54 Complete the reaction:



A. 2-Phenylethanol ($C_6H_5CH_2CH_2OH$) + CH_3OH

B. 2-Phenylethylamine

C. Phenylacetic acid + CH_3OH

D. Phenylacetaldehyde + CH_3OH

Answers · Lecture 11

11.1 Structure & Nomenclature of Carboxylic Acids

Q1 **B. R–COOH — a carbonyl (C=O) and a hydroxyl (–OH) on the same carbon**

Carboxylic acids contain the **carboxyl group** –COOH (also written –CO_2H): carb**o**nyl + hydr**oxyl**.

Q2 **A. -oic acid**

Choose the longest chain that contains the COOH, drop the final -e of the alkane name and add **-oic acid** (e.g. propane → propanoic acid).

Q3 **D. Always C1**

The carboxyl carbon is always C1, so no number is needed for the COOH. Other groups get numbers from it.

Q4 **C. α (alpha)**

Common names use Greek letters: the carbon next to COOH = α, then β, γ, δ … (IUPAC C2 = α).

Q5 **D. oxo**

The carboxylic acid has the highest priority, so it gets the suffix (-oic acid); the ketone becomes **oxo**, OH becomes hydroxy, NH_2 becomes amino.

Q6 **C. 3-Chlorocyclopentanecarboxylic acid**

COOH on a ring → **cycloalkanecarboxylic acid**. The ring carbon with COOH = C1, Cl on C3 → **3-chlorocyclopentanecarboxylic acid**.

Q7 **C. 4-Bromobenzoic acid (p-bromobenzoic acid)**

Benzene + COOH = benzoic acid; the COOH carbon is C1, Br is opposite → C4 (para) → **4-bromobenzoic acid**.

Q8 **B. 2-Butenoic acid**

4 carbons with a C=C between C2 and C3 → but**en**oic acid, numbered from the COOH → **2-butenoic acid**.

Q9 **C. 2-Aminopropanoic acid**

3-carbon chain, COOH = C1, NH_2 on C2 → **2-aminopropanoic acid** (the amino acid alanine).

Q10 **A. 3-Oxopentanoic acid**

5 carbons, COOH = C1, ketone C=O on C3 → **3-oxopentanoic acid**.

11.2 Physical Properties

Q11 **C. Carboxylic acids boil higher**

Carboxylic acids boil **higher** than alcohols of similar size because they form strong hydrogen bonds — two molecules join as a **hydrogen-bonded dimer**.

Q12 **A. They form hydrogen-bonded dimers (two H-bonds between two molecules)**

Each COOH has an H-bond donor (O–H) and an acceptor (C=O), so two molecules hold each other with two H-bonds (dimer).

Q13 **D. Small acids (about 1–4 carbons) are water-soluble because of H-bonding; solubility decreases as the carbon chain gets longer**

The COOH group H-bonds with water, so small acids mix with water. A long non-polar R group makes the acid less soluble.

Q14 **A. $CH_3OCOCH_2COOCH_3$ < $CH_3OCOCH_2CH_2COOH$ < $HOOC(CH_2)_3COOH$**

Same molecular weight, so H-bonding decides: the diester has **no O–H** (lowest bp) < the ester-acid has **one COOH** < the diacid has **two COOH** (highest bp).

11.3 Acidity of Carboxylic Acids

Q15 **C. Carboxylic acid > phenol > alcohol**

pKa ≈ 4–5 (carboxylic acid) · ≈ 10 (phenol) · ≈ 16 (alcohol). Lower pKa = stronger acid.

Q16 **C. The carboxylate ion ($RCOO^-$) is stabilized by resonance — the negative charge is shared equally by two oxygens**

After losing H^+, the negative charge of $RCOO^-$ is delocalized over both oxygens (two equivalent resonance structures). A more stable anion = a stronger acid.

Q17 **C. $NaHCO_3$ (sodium bicarbonate, weak base) — the acid gives CO_2 bubbles**

$RCOOH + NaHCO_3 \rightarrow RCOO^-Na^+ + H_2O + CO_2\uparrow$. Phenols are too weak to react with $NaHCO_3$, but they do react with the strong base NaOH.

Q18 **A. NaOH (strong base) reacts with both carboxylic acids and phenols to form water-soluble salts**

$RCOOH + NaOH \rightarrow RCOO^-Na^+ + H_2O$ · $ArOH + NaOH \rightarrow ArO^-Na^+ + H_2O$. Alcohols are too weak to react with aqueous NaOH.

Q19 **D. Electron-withdrawing groups increase acidity; the closer they are to the COOH (and the more there are), the stronger the acid**

Inductive effect: EWG (NO_2, F, Cl, Br, I) pull electrons and stabilize $RCOO^-$ → stronger acid. The more electronegative the group, the stronger the effect (F > Cl > Br > I). The effect gets weaker with distance (2-Cl 2.86 → 3-Cl 4.05 → 4-Cl 4.52). Alkyl groups donate electrons → weaker acid (HCOOH 3.75 vs CH_3COOH 4.76).

Q20 **B. FCH_2COOH**

pKa: F 2.59 < Cl 2.87 < Br 2.90 < I 3.17. F is the most electronegative, so FCH_2COOH is the strongest.

Q21 **B. The CH_3 group donates electrons and destabilizes the carboxylate ion**

Alkyl groups are electron-donating; they increase the negative charge on COO^- → less stable → weaker acid.

Q22 **B. $ClCH_2CH_2COOH < CH_3CHClCOOH < CH_3CHFCOOH < CH_2FCHFCOOH$**

Least: Cl far from COOH (C3) < Cl on C2 < F on C2 (F more electronegative) < two F atoms (most).

Q23 **B. 4-Oxopentanoic acid < 3-oxopentanoic acid < 2-oxopentanoic acid**

The C=O is electron-withdrawing; the closer it is to the COOH, the stronger the acid: C=O on C4 (least) < C3 < C2 (most).

11.4 Preparation of Carboxylic Acids

Q24 **A. Oxidation of alkylbenzenes, oxidation of 1° alcohols, hydrolysis of nitriles**

The 3 methods: (1) oxidation of alkylbenzenes, (2) oxidation of primary alcohols, (3) hydrolysis of nitriles.

Q25 **A. Benzoic acid (C_6H_5COOH)**

Strong oxidizing agents ($KMnO_4$ or $K_2Cr_2O_7/H^+$) oxidize the alkyl side chain on a benzene ring to –COOH. The side chain needs a benzylic H.

Q26 **D. A carboxylic acid**

R–CH_2OH → [R–CHO] → R–COOH. A strong oxidant continues past the aldehyde to the acid.

Q27 **A. R–COOH + NH_4^+**

R–C≡N + 2 H_2O (H^+, heat) → R–COOH + NH_4^+. (Base hydrolysis gives $RCOO^-$, then H_3O^+ gives RCOOH.)

11.5 Carboxylic Acid Derivatives: Structure & Names

Q28 **D. Acid chlorides (RCOCl), acid anhydrides (RCO–O–COR), esters (RCOOR'), amides ($RCONH_2$)**

In each derivative the –OH of the acid is replaced: by Cl (acid chloride), by –O–COR (anhydride), by –OR' (ester), by –NH_2/–NHR/–NR_2 (amide).

Q29 **D. Change -ic acid to -yl chloride (propanoic acid → propanoyl chloride)**

Propanoic acid → **propanoyl chloride**; benzoic acid → **benzoyl chloride**.

Q30 **D. Replace the word acid with anhydride: propanoic anhydride, benzoic anhydride**

Two acid molecules joined with loss of water → name the acid and change **acid** to **anhydride**.

Q31 **A. Ethyl propanoate and ethyl benzoate**

Ester = **alkyl group on O** first (ethyl) + acid name with **-ic acid** → **-ate** (propanoate, benzoate).

Q32 **C. Propanamide and benzamide**

Amide: change **-oic acid / -ic acid** → **-amide**: propanoic acid → propanamide, benzoic acid → benzamide.

11.6 Preparation of Acid Derivatives

Q33 **B. $SOCl_2$ (thionyl chloride) or PCl_5**

RCOOH + $SOCl_2$ → RCOCl + SO_2 + HCl (or with PCl_5 / PCl_3).

Q34 **C. An acid chloride + a carboxylate salt (RCOCl + $RCOO^-Na^+$)**

RCOCl + $R'COO^-Na^+$ → RCO–O–COR' + NaCl. (Anhydrides can also form by removing water from two COOH groups with heat.)

Q35 **C. RCOOH + R'OH with an acid catalyst (H^+) ⇌ RCOOR' + H_2O**

Esters are made from a carboxylic acid + an alcohol with H^+ (reversible), or from an acid chloride + an alcohol.

Q36 **B. An acid chloride + ammonia or an amine**

RCOCl + 2 NH_3 → $RCONH_2$ + NH_4Cl (with RNH_2 → RCONHR; with R_2NH → $RCONR_2$).

11.7 Nucleophilic Acyl Substitution

Q37 **B. R–CO–L + Nu^- → R–CO–Nu + L^-**

The group L attached to the acyl group (RC=O) is replaced by the nucleophile. It goes through a tetrahedral intermediate (addition, then elimination of L).

Q38 **D. Acid chloride > acid anhydride > ester > amide**

The better the leaving group (Cl^- is best, NH_2^- is worst), the more reactive the derivative. A more reactive derivative can be changed into a less reactive one, not the reverse.

Q39 **D. Carboxylic acid + HCl**

RCOCl + H_2O → RCOOH + HCl (hydrolysis).

Q40 **C. Ester + HCl**

RCOCl + R'OH → RCOOR' + HCl (alcoholysis).

Q41 **B. Amide (RCONHR') + HCl**

RCOCl + $R'NH_2$ → RCONHR' + HCl (aminolysis; a second amine removes the HCl).

Q42 **D. Two carboxylic acids**

$(RCO)_2O$ + H_2O → 2 RCOOH.

Q43 **A. Ester + carboxylic acid**

$(RCO)_2O$ + R'OH → RCOOR' + RCOOH.

Q44 **C. Amide + carboxylic acid**

$(RCO)_2O$ + $R'NH_2$ → RCONHR' + RCOOH.

Q45 **A. Carboxylic acid (or carboxylate in base) + alcohol**

RCOOR' + H_2O (H^+) ⇌ RCOOH + R'OH. In base (saponification): RCOOR' + NaOH → $RCOO^-Na^+$ + R'OH.

Q46 **D. A new ester + R'OH (transesterification)**

RCOOR' + R''OH (H^+ or base) ⇌ RCOOR'' + R'OH.

Q47 **A. Amide + alcohol (R'OH)**

RCOOR' + $R''NH_2$ → RCONHR'' + R'OH (ester → less reactive amide).

Q48 **B. Carboxylic acid (or its salt) + amine / NH_3**

Amides are the least reactive; they hydrolyze only with strong acid or base and heat: $RCONH_2$ + H_2O → RCOOH + NH_4^+ (acid) or $RCOO^-$ + NH_3 (base).

Q49 **A. No reaction — amides are the least reactive derivative**

An amide cannot be converted into a more reactive derivative (ester) and does not normally react with amines.

11.8 Reactions with Grignard Reagents & $LiAlH_4$

Q50 **B. A 3° alcohol with two R'' groups + R'OH**

The first R''MgX substitutes the OR' (gives a ketone), the second adds to the ketone → after H_3O^+: R–$C(OH)R''_2$ (3° alcohol) + R'OH.

Q51 **B. A 1° alcohol (RCH_2OH) + R'OH**

$LiAlH_4$ (strong reducing agent) reduces the ester to two alcohols: RCH_2OH (from the acyl part) + R'OH.

Q52 **B. An amine, RCH_2NR_2**

$LiAlH_4$ changes the C=O of the amide into CH_2; the nitrogen stays → amine.

Q53 **D. 2-Phenyl-2-propanol, $C_6H_5C(OH)(CH_3)_2$ + CH_3CH_2OH**

Ethyl benzoate + 2 CH_3MgBr → the two CH_3 groups add to the carbonyl C, the OCH_2CH_3 leaves as ethanol → 3° alcohol **$C_6H_5C(OH)(CH_3)_2$**.

Q54 **A. 2-Phenylethanol ($C_6H_5CH_2CH_2OH$) + CH_3OH**

Methyl phenylacetate + $LiAlH_4$ → the acyl part becomes the 1° alcohol **$C_6H_5CH_2CH_2OH$**; the OCH_3 part becomes methanol.

12 Alkyl Halides

In this lecture

1. Types, Uses & Classification 9 questions
2. Physical Properties 3 questions
3. Nomenclature of Alkyl Halides 14 questions
4. Preparation of Alkyl Halides 5 questions
5. Reactions: Substitution & Elimination 8 questions
6. Nucleophilic Substitution Practice 14 questions
7. Safety of Organohalogen Compounds 1 questions

12.1 Types, Uses & Classification

Q1 Which halogens are found in alkyl halides (organohalogen compounds)?

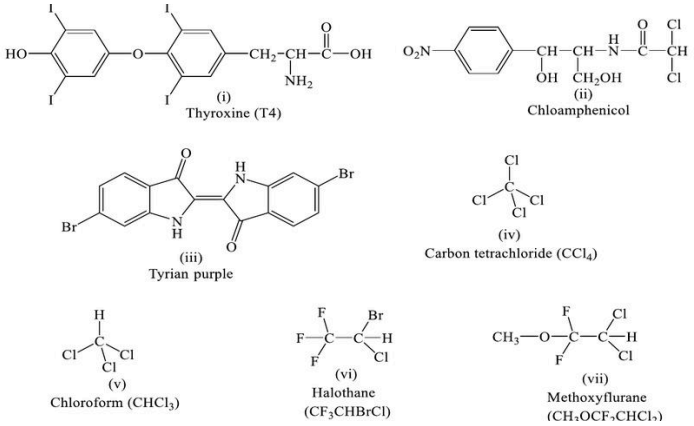
A. F, Cl, Br and I

B. Cl, Br, S and P

C. F, Cl, O and N

D. Only Cl and Br

Q2 Which compounds are organic solvents that may cause liver damage?



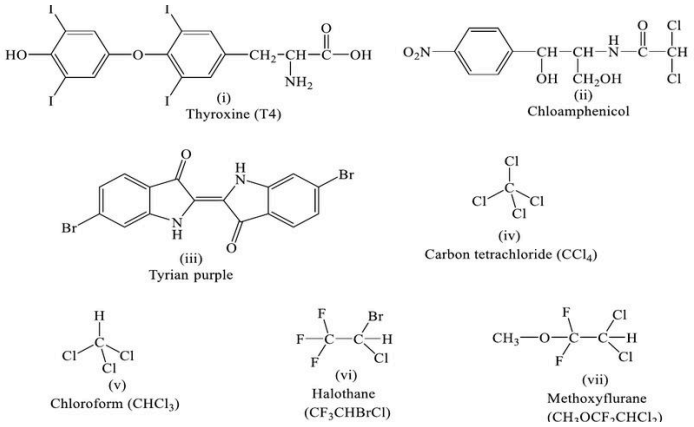
A. Halothane and methoxyflurane

B. Tyrian purple

C. Thyroxine and chloramphenicol

D. Carbon tetrachloride (CCl_4) and chloroform ($CHCl_3$)

Q3 Which alkyl halides are safe and useful as anesthetics?



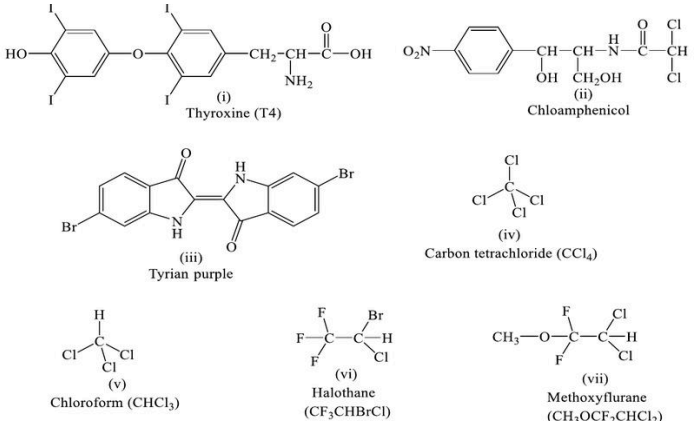
A. Halothane ($CF_3CHBrCl$) and methoxyflurane ($CH_3OCF_2CHCl_2$)

B. CCl_4 and $CHCl_3$

C. Thyroxine

D. Chloramphenicol

Q4 Match: a naturally occurring dye / a thyroid hormone component / an antibiotic from a soil bacterium.



A. Chloramphenicol / Tyrian purple / thyroxine

B. Halothane / thyroxine / CCl_4

C. Thyroxine / chloramphenicol / Tyrian purple

D. Tyrian purple / thyroxine (T4) / chloramphenicol

Q5 In which halide is the halogen bonded to an sp^2 carbon?

A. Allylic halide (CH_2=CH–CH_2–Cl)

B. Methyl halide (CH_3–Cl)

C. Benzylic halide (C_6H_5–CH_2–Cl)

D. Aryl halide (C_6H_5–Cl) and vinylic halide (CH_2=CH–Cl)

Q6 The halogen atom in an allylic halide is attached to an:

A. sp carbon

B. sp^3 carbon

C. sp^2 carbon

D. None of the above

Q7 A benzylic halide is:

A. X on a carbon attached to a benzene ring, e.g. $C_6H_5CH_2Cl$

B. X on a CH_3 group only

C. X on a C=C carbon

D. X directly on the benzene ring, e.g. C_6H_5Cl

Q8 How are alkyl halides classified as 1°, 2° and 3°?

A. By the number of halogen atoms

B. By the number of carbon atoms bonded to the carbon that carries the halogen (1, 2 or 3)

C. By the length of the chain

D. By the type of halogen

Q9 $(CH_3)_3C–Br$ is a:

A. 2° alkyl halide

B. 1° alkyl halide

C. Methyl halide

D. 3° alkyl halide

12.2 Physical Properties

Q10 From the table, which compound has the highest and which the lowest boiling point?

R–X	Cl	Br	I
CH_3–X	−24 (gas)	5 (gas)	43 (liquid)
CH_3CH_2–X	12.5 (gas)	38 (liquid)	72 (liquid)
CH_2X_2	40 (liquid)	99 (liquid)	180 (liquid)
CHX_3	61 (liquid)	151 (liquid)	sublimes (solid)
CX_4	77 (liquid)	189 (liquid)	sublimes (solid)
CH_2=CH–X	−14 (gas)	14 (gas)	56 (gas)
Phenyl–X	131 (liquid)	156 (liquid)	188 (liquid)

b.p. in °C (physical state)

A. Highest: CH_3Cl; lowest: CBr_4

B. Highest: CBr_4 (189 °C) — and CHI_3/CI_4 are solids; lowest: CH_3Cl (−24 °C)

C. Highest: CH_2=CHCl; lowest: phenyl iodide

D. All have the same boiling point

Q11 For the same alkyl group, the boiling point order is:

A. R–I > R–Br > R–Cl

B. They are all equal

C. R–Br > R–I > R–Cl

D. R–Cl > R–Br > R–I

Q12 Are organohalogen compounds water-soluble?

A. Only the iodides are soluble

B. Yes — they are ionic

C. Yes — they form strong hydrogen bonds with water

D. No — they cannot form hydrogen bonds with water, so they are insoluble (but dissolve in organic solvents)

12.3 Nomenclature of Alkyl Halides

Q13 In IUPAC names, halogens are named as:

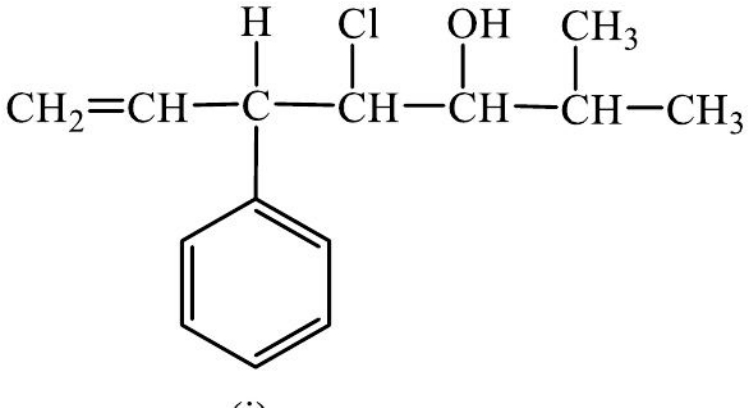
A. Suffixes: -fluoride, -chloride

B. They are not named

C. Suffixes: -ol

D. Prefixes: fluoro-, chloro-, bromo-, iodo-

Q14 Name this compound:



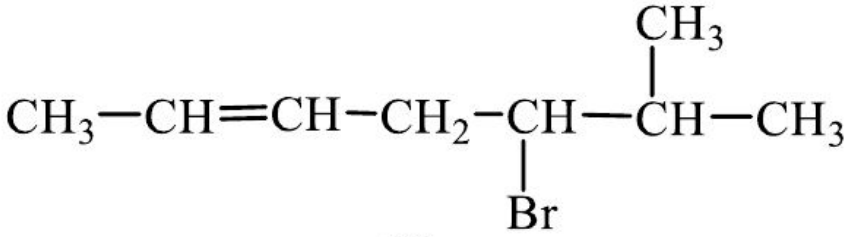
A. 5-Chloro-2-methyl-6-hepten-3-ol

B. 4-Chloro-2-methyl-5-phenylhept-6-en-3-ol

C. 4-Chloro-6-methyl-3-phenylhept-1-en-5-ol

D. 4-Chloro-2-methyl-5-phenylhept-6-en-3-one

Q15 Name this compound:



(ii)

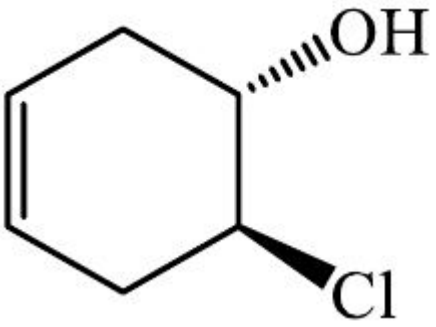
A. 2-Methyl-3-bromohept-5-ene

B. 5-Bromo-6-methylhept-2-ene

C. 5-Bromo-6-methylhept-3-ene

D. 3-Bromo-2-methylhept-5-ene

Q16 Name this compound:



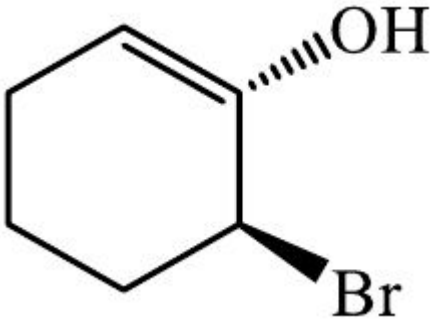
A. (1R,6S)-1-Chlorocyclohex-3-en-6-ol

B. (1R,2R)-2-Chlorocyclohex-4-en-1-ol

C. (1S,6R)-6-Chlorocyclohex-3-en-1-ol

D. (1S,6S)-6-Chlorocyclohex-3-en-1-ol

Q17 Name this compound:



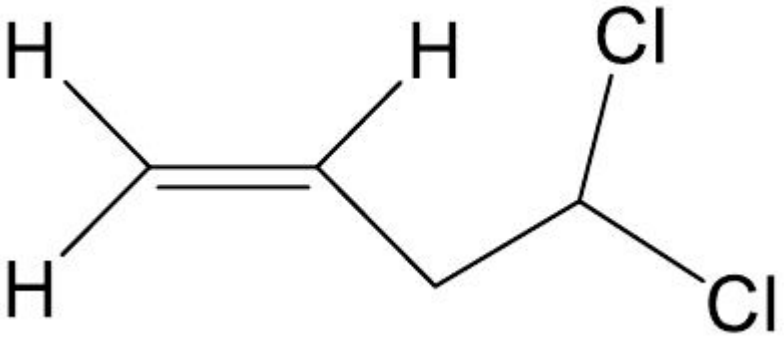
A. (S)-2-Bromocyclohex-6-en-1-ol

B. (R)-6-Bromocyclohex-1-en-1-ol

C. (S)-6-Bromocyclohex-1-en-1-ol

D. (R)-1-Bromocyclohex-2-en-3-ol

Q18 The correct IUPAC name for this compound is:



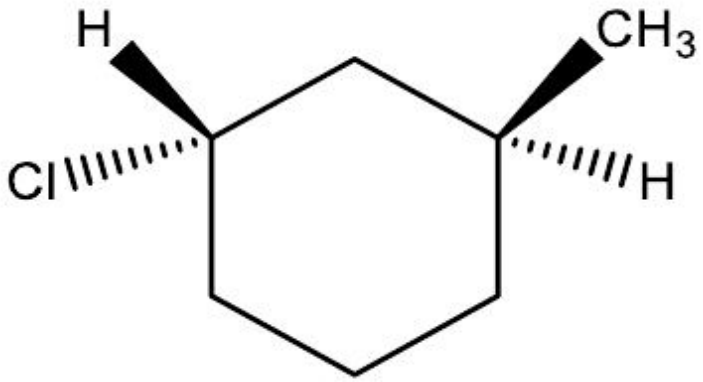
A. 1,1-Dichlorobut-1-ene

B. 1,1-Dichlorobut-3-ene

C. 1,1-Dichlorobutene

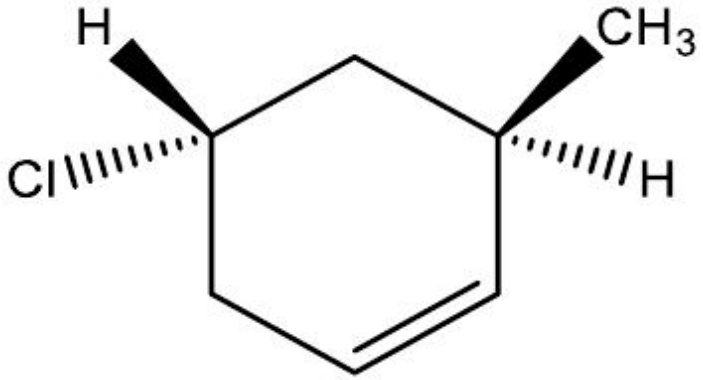
D. 4,4-Dichloro-1-butene

Q19 The correct IUPAC name for this compound is:



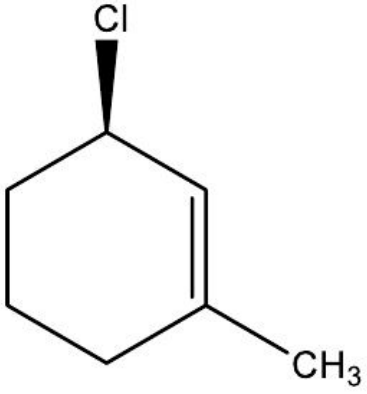
A. (1S,3S)-1-Chloro-3-methylcyclohexane
B. (1S,3R)-1-Chloro-3-methylcyclohexane
C. (1R,3S)-1-Chloro-3-methylcyclohexane
D. (1R,3R)-1-Chloro-3-methylcyclohexane

Q20 The correct IUPAC name for this compound is:



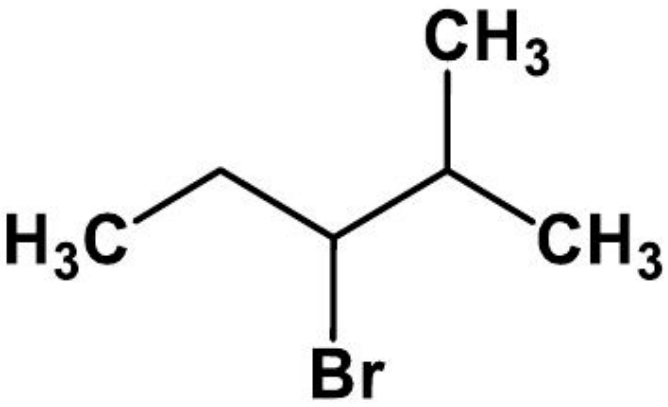
A. (4R,6S)-4-Chloro-6-methylcyclohex-1-ene
B. (4S,6R)-4-Chloro-6-methylcyclohex-1-ene
C. (3R,5S)-5-Chloro-3-methylcyclohex-1-ene
D. (3S,5R)-5-Chloro-3-methylcyclohex-1-ene

Q21 What is the IUPAC name of this compound?



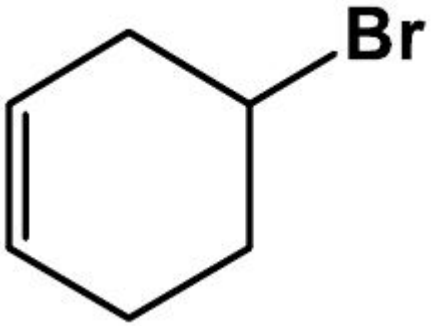
A. (R)-3-Chloro-1-methylcyclohexene
B. (R)-1-Chloro-3-methyl-2-cyclohexene
C. (S)-3-Chloro-1-methylcyclohexene
D. (S)-1-Chloro-3-methyl-2-cyclohexene

Q22 Name this compound:



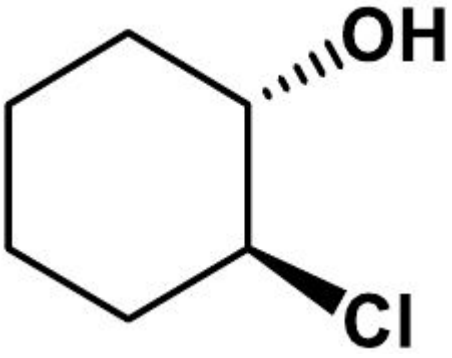
A. 3-Bromo-2-methylpentane
B. 2-Methyl-3-bromopentane
C. 3-Bromo-2-ethylbutane
D. 3-Bromo-4-methylpentane

Q23 Name this compound:



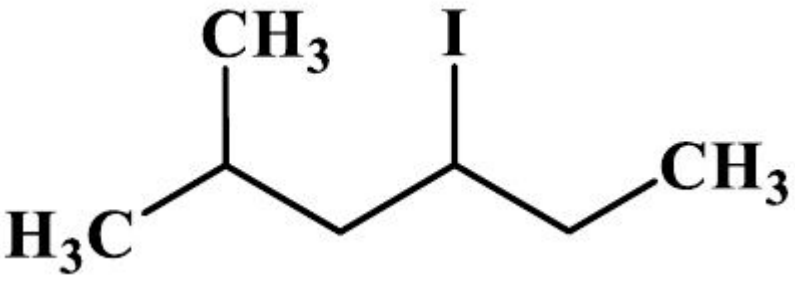
A. 5-Bromocyclohexene
B. 3-Bromocyclohexene
C. 1-Bromocyclohex-3-ene
D. 4-Bromocyclohexene

Q24 Name this compound:



A. (1S,2S)-2-Chlorocyclohexan-1-ol
B. (1R,2S)-1-Chlorocyclohexan-2-ol
C. (1S,2R)-2-Chlorocyclohexan-1-ol
D. (1R,2R)-2-Chlorocyclohexan-1-ol

Q25 Name this compound:



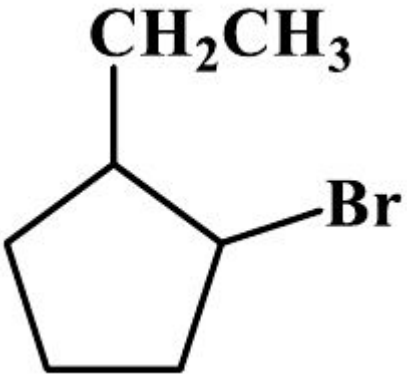
A. 3-Iodo-5-methylhexane

B. 4-Iodo-2-ethylpentane

C. 4-Iodo-2-methylhexane

D. 2-Methyl-4-iodohexane

Q26 Name this compound:



A. 1-Bromo-2-ethylcyclopentane

B. 1-Bromo-2-ethylcyclohexane

C. 2-Bromo-1-ethylcyclopentane

D. 1-Ethyl-2-bromocyclopentane

12.4 Preparation of Alkyl Halides

Q27 Which reactions of alkenes and alcohols prepare alkyl halides?

A. Oxidation of alcohols with $KMnO_4$

B. Hydrogenation of alkenes

C. Addition of HX or X_2 to alkenes, and substitution of the OH of alcohols (HX, $SOCl_2$, PBr_3)

D. Ozonolysis of alkenes

Q28 $CH_3CH{=}CH_2$ + HBr gives (major product):

A. $CH_3CH_2CH_3$

B. $CH_3CH_2CH_2Br$ (1-bromopropane)

C. $CH_3CHBrCH_3$ (2-bromopropane) — Markovnikov

D. $CH_3CHBrCH_2Br$

Q29 $CH_2=CH_2 + Br_2$ gives:

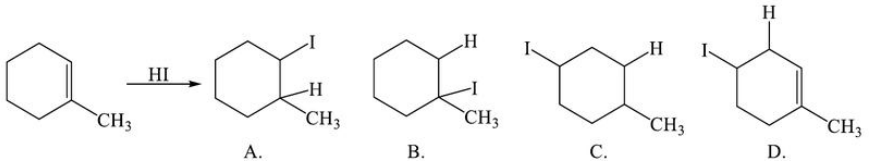
A. CH_3CH_2Br

B. $BrCH_2CH_2Br$ (1,2-dibromoethane, a vicinal dihalide)

C. No reaction

D. $CH_2=CHBr$

Q30 Predict the product of this reaction:



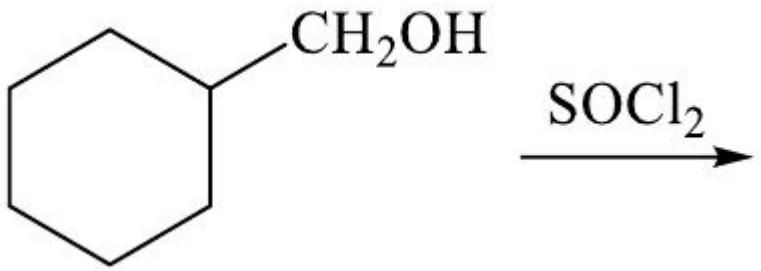
A. 2-Iodo-1-methylcyclohexane (structure A)

B. 4-Iodo-1-methylcyclohexene (structure D)

C. 1-Iodo-1-methylcyclohexane (structure B)

D. 1-Iodo-4-methylcyclohexane (structure C)

Q31 Predict the product of this reaction:



A. 2-Chloro-2-methylcyclohexane

B. Methylcyclohexene

C. (Chloromethyl)cyclohexane

D. 3-Chloro-1-methylcyclohexene

12.5 Reactions: Substitution & Elimination

Q32 Which two types of reactions do alkyl halides undergo?

A. Addition and polymerization

B. Nucleophilic substitution and elimination

C. Hydration and ozonolysis

D. Oxidation and reduction

Q33 In an elimination reaction of an alkyl halide, the product is:

A. An alcohol

B. A longer alkyl halide

C. An alkene (H and X are removed from neighbouring carbons)

D. An ether

Q34 What is a nucleophile?

A. An electron-rich species (anion or neutral molecule with a lone pair) that donates an electron pair to an electron-poor carbon

B. A positively charged ion such as H^+

C. A free radical

D. An electron-poor species that accepts electrons

Q35 Which are oxygen nucleophiles?

A. OH^-, RO^- (alkoxide), H_2O, ROH, $RCOO^-$ (carboxylate)

B. NH_3, RNH_2

C. CN^-, $RC{\equiv}C^-$

D. HS^-, RS^-

Q36 Which are nitrogen nucleophiles?

A. CN^-, $RC{\equiv}C^-$

B. OH^-, RO^-

C. I^-, Br^-, Cl^-

D. NH_3, RNH_2, R_2NH, R_3N, N_3^- (azide)

Q37 Which are sulphur nucleophiles?

A. OH^-, H_2O

B. HS^-, RS^- (thiolate), H_2S, RSH

C. CN^-

D. NH_3, N_3^-

Q38 Which are carbon nucleophiles?

A. NH_3 and RNH_2

B. Cl^- and Br^-

C. OH^- and RO^-

D. CN^- (cyanide) and $RC{\equiv}C^-$ (acetylide)

Q39 Which are halogen nucleophiles?

A. HCl gas only

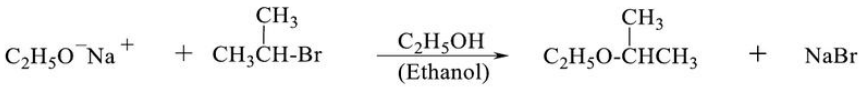
B. X^+ ions

C. I^-, Br^-, Cl^- (halide ions)

D. F_2, Cl_2

12.6 Nucleophilic Substitution Practice

Q40 In this reaction, identify the alkyl halide, the nucleophile and the solvent:



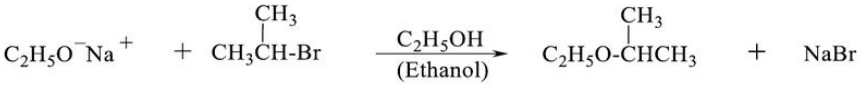
A. 3° alkyl halide; nucleophile = Br^-; solvent = ethanol

B. 2° alkyl halide; nucleophile = ethanol; solvent = NaBr

C. 2° alkyl halide (2-bromopropane); nucleophile = ethoxide ($C_2H_5O^-$); solvent = ethanol

D. 1° alkyl halide; nucleophile = Na^+; solvent = water

Q41 In the same reaction, identify the leaving group, the product and the by-product:



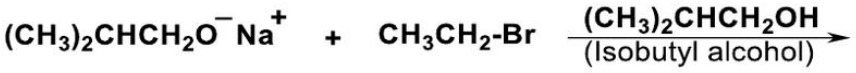
A. Leaving group = OH^-; product = alcohol; by-product = NaBr

B. Leaving group = Br^-; product = $C_2H_5OCH(CH_3)_2$ (2-ethoxypropane, an ether); by-product = NaBr

C. Leaving group = $C_2H_5O^-$; product = NaBr; by-product = ether

D. Leaving group = Na^+; product = propene; by-product = ethanol

Q42 Predict the product:



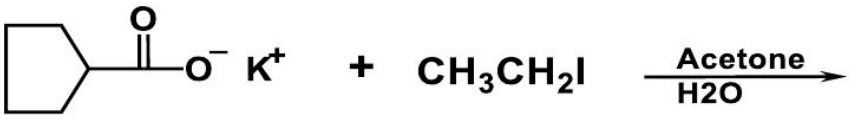
A. $CH_2{=}CH_2$

B. $(CH_3)_2CHCH_2{-}O{-}CH_2CH_3$ (ethyl isobutyl ether) + NaBr

C. $(CH_3)_2CHCH_2OH + CH_3CH_2Br$

D. $(CH_3)_2CHCH_2{-}CH_2CH_3$

Q43 Predict the product:



A. Ethyl cyclobutanecarboxylate (an ester) + KI

B. Cyclobutanecarboxylic acid + ethane

C. Ethyl cyclobutyl ketone

D. Cyclobutylmethanol

Q44 Predict the product:



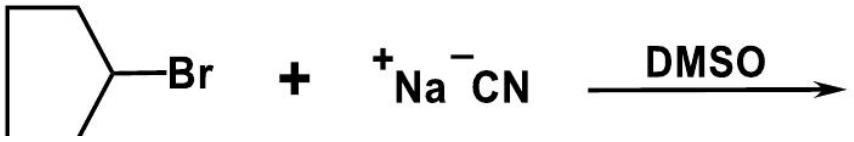
A. Cyclobutene

B. Cyclobutanethiol (cyclobutyl–SH) + KBr

C. Cyclobutanol

D. Dicyclobutyl sulfide only

Q45 Predict the product:



A. Cyclobutene

B. Cyclobutanol

C. Cyclobutylamine

D. Cyclobutanecarbonitrile (cyclobutyl–C≡N) + NaBr

Q46 Predict the product:



A. $CH_3CH_2C{\equiv}CCH_3$ (2-pentyne) + NaBr

B. $CH_3CH_2C{\equiv}CH$ + CH_3Na

C. $CH_3CH_2C{\equiv}C–Br$

D. $CH_3CH_2CH{=}CHCH_3$

Q47 Predict the two products (??):



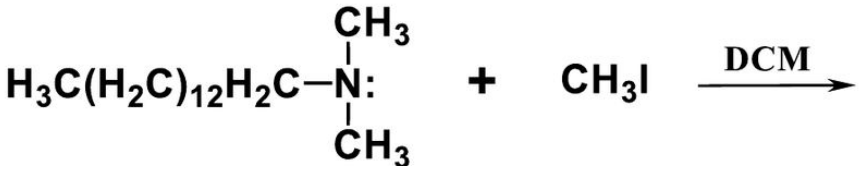
A. $(CH_3)_2N^+H–CH_2CH_3$ I^- (ammonium salt), then with base → $(CH_3)_2N–CH_2CH_3$ (a 3° amine)

B. CH_3NH_2 + CH_3CH_2I

C. No reaction

D. $CH_3–NH–CH_3$ + CH_3CH_2I

Q48 Predict the product:



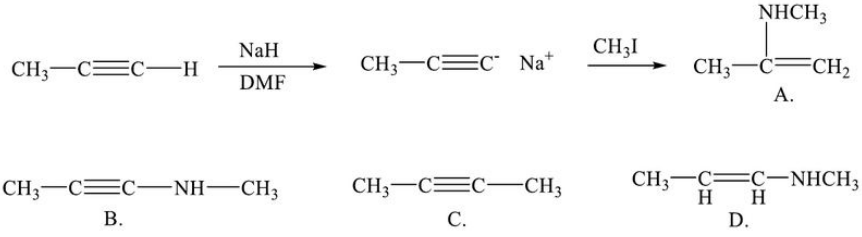
A. $CH_3(CH_2)_{13}I$ + $(CH_3)_3N$

B. $CH_3(CH_2)_{13}N^+(CH_3)_3$ I^- — a quaternary ammonium salt

C. $CH_3(CH_2)_{13}N(CH_3)_2$ + CH_3I (no reaction)

D. An alkene

Q49 The major product of this reaction will be:



A. $CH_3C(NHCH_3){=}CH_2$ (structure A)

B. $CH_3CH{=}CH–NHCH_3$ (structure D)

C. $CH_3C{\equiv}CCH_3$, 2-butyne (structure C)

D. $CH_3C{\equiv}C–NHCH_3$ (structure B)

Q50 Which alkyl halide + nucleophile give $CH_3OCH_2CH_2CH_2CH_3$?

A. CH_3Br + $CH_3CH_2CH_2CH_3$

B. $CH_3CH_2CH_2CH_2OH$ + CH_3OH

C. $CH_3CH_2CH_2CH_2Br$ + NaOH

D. $CH_3CH_2CH_2CH_2Br$ + $CH_3O^-Na^+$

Q51 Which alkyl halide + nucleophile give $HC{\equiv}CCH_2CH_2CH_3$?

A. $CH_3CH_2CH_2Br$ + NaOH

B. $CH_3CH_2CH_2Br$ + $HC{\equiv}C^-Na^+$

C. $CH_3CH_2CH_2OH$ + $HC{\equiv}CH$

D. $HC{\equiv}CBr$ + $CH_3CH_2CH_3$

Q52 Which alkyl halide + nucleophile give $CH_3CH_2CH_2NH_2$?

A. $CH_3CH_2CH_2Br$ + NaOH

B. $CH_3CH_2CH_3$ + NH_3

C. $CH_3CH_2CH_2Br$ + NH_3 (excess)

D. $CH_3CH_2CH_2OH$ + HCl

Q53 Which alkyl halide + nucleophile give $CH_3CH_2CH_2SCH_2CH_2CH_3$?

A. $CH_3CH_2CH_2OH$ + H_2S

B. $CH_3CH_2CH_2Br$ + $CH_3CH_2CH_2S^-Na^+$

C. $CH_3CH_2CH_2Br$ + NaSH only

D. $CH_3CH_2CH_3$ + S

12.7 Safety of Organohalogen Compounds

Q54 Are all organohalogen compounds safe for oral use?

A. Yes, all are safe

B. Yes, because they are insoluble in water

C. No — e.g. CCl_4 and $CHCl_3$ are toxic and can cause liver damage

D. No — only the fluorides are toxic

Answers · Lecture 12

12.1 Types, Uses & Classification

Q1 **A. F, Cl, Br and I**

The four halogens: **fluorine (F), chlorine (Cl), bromine (Br), iodine (I)**. General formula R–X (X = F, Cl, Br, I).

Q2 **D. Carbon tetrachloride (CCl_4) and chloroform ($CHCl_3$)**

CCl_4 (iv) and $CHCl_3$ (v) are good organic solvents, but they are toxic to the liver.

Q3 **A. Halothane ($CF_3CHBrCl$) and methoxyflurane ($CH_3OCF_2CHCl_2$)**

Halothane (vi) and methoxyflurane (vii) are inhalation anesthetics.

Q4 **D. Tyrian purple / thyroxine (T4) / chloramphenicol**

Tyrian purple (iii) = natural dye (contains Br) · **thyroxine T4** (i) = thyroid hormone (contains I) · **chloramphenicol** (ii) = antibiotic from a soil bacterium (Streptomyces; contains Cl).

Q5 **D. Aryl halide (C_6H_5–Cl) and vinylic halide (CH_2=CH–Cl)**

Methyl CH_3X, alkyl (e.g. CH_3CH_2Br), allylic CH_2=$CHCH_2X$ and benzylic $C_6H_5CH_2X$ → X on an **sp^3** carbon. Aryl C_6H_5X and vinylic CH_2=CHX → X on an **sp^2** carbon.

Q6 **B. sp^3 carbon**

Allylic = the carbon **next to** a C=C (CH_2=CH–CH_2–X). That carbon is sp^3.

Q7 **A. X on a carbon attached to a benzene ring, e.g. $C_6H_5CH_2Cl$**

Benzylic = C_6H_5–CH_2–X (sp^3). Aryl = C_6H_5–X (sp^2).

Q8 **B. By the number of carbon atoms bonded to the carbon that carries the halogen (1, 2 or 3)**

1°: RCH_2–X (one C attached) · 2°: R_2CH–X (two C) · 3°: R_3C–X (three C).

Q9 **D. 3° alkyl halide**

The C–Br carbon is bonded to three CH_3 groups → tertiary.

12.2 Physical Properties

Q10 **B. Highest: CBr_4 (189 °C) — and CHI_3/CI_4 are solids; lowest: CH_3Cl (−24 °C)**

Boiling (and melting) point increases with **molecular weight**: bigger halogen (Cl < Br < I) and more halogens (CH_3X < CH_2X_2 < CHX_3 < CX_4). CHI_3 and CI_4 are even solids. CH_3Cl is the smallest → gas, lowest bp.

Q11 **A. R–I > R–Br > R–Cl**

Example CH_3CH_2–X: Cl 12.5 °C < Br 38 °C < I 72 °C. Larger atom → more polarizable → stronger van der Waals forces.

Q12 **D. No — they cannot form hydrogen bonds with water, so they are insoluble (but dissolve in organic solvents)**

Even though the C–X bond is polar, alkyl halides have no O–H or N–H, so they cannot H-bond with water → insoluble in water, soluble in non-polar organic solvents.

12.3 Nomenclature of Alkyl Halides

Q13 **D. Prefixes: fluoro-, chloro-, bromo-, iodo-**

Halogens are always prefixes; they get numbers like alkyl groups and are listed in alphabetical order. Suffix groups (–OH, C=C …) decide the numbering.

Q14 **B. 4-Chloro-2-methyl-5-phenylhept-6-en-3-ol**

Longest chain with OH and C=C = 7 C (**hept**). OH is the suffix (**-ol**) → number from the right so OH = C3. C=C at C6 (**hept-6-en**). Substituents: Cl on C4, CH_3 on C2, phenyl on C5 → alphabetical: **4-chloro-2-methyl-5-phenylhept-6-en-3-ol**.

Q15 **B. 5-Bromo-6-methylhept-2-ene**

Chain = 7 C. The C=C (suffix -ene) must get the lowest number → number from the left: C=C at C2. Br on C5, CH_3 on C6 → **5-bromo-6-methylhept-2-ene**.

Q16 **D. (1S,6S)-6-Chlorocyclohex-3-en-1-ol**

OH carbon = C1. Number towards the C=C so it gets C3–C4 (not 4–5); then Cl is on C6 → 6-chlorocyclohex-3-en-1-ol. OH (hashed) and Cl (wedge) are trans. C1: OH > C6 > C2 > H, H points to the viewer → **S**. C6: Cl > C1 > C5 > H, H away → **S**.

Q17 **C. (S)-6-Bromocyclohex-1-en-1-ol**

OH is on a C=C carbon: OH carbon = C1, its double-bond partner = C2 (cyclohex-1-en-1-ol). Br is on the other neighbour, C6. At C6: Br > C1 (=C–O) > C5 > H; Br is wedged (H away) → **S**.

Q18 **D. 4,4-Dichloro-1-butene**

CH_2=CH–CH_2–$CHCl_2$. The C=C (suffix -ene) gets the lowest number → C1, so the Cl atoms are on C4 → **4,4-dichloro-1-butene (4,4-dichlorobut-1-ene)**.

Q19 **A. (1S,3S)-1-Chloro-3-methylcyclohexane**

No suffix group, so Cl gets C1 by alphabetical order (chloro before methyl). C1: Cl > C2 (towards C3) > C6 > H; H is wedged (toward you) → reverse → **S**. C3: C2 (towards C–Cl) > C4 > CH_3 > H; H hashed (away) → counter-clockwise → **S**.

Q20 **C. (3R,5S)-5-Chloro-3-methylcyclohex-1-ene**

C=C = C1–C2, numbered so the substituents get {3,5} (not {4,6}) → 5-chloro-3-methylcyclohex-1-ene. C3: C2 (=CH) > C4 > CH_3 > H, H away → **R**. C5: Cl > C4 > C6 > H, H toward you → **S**.

Q21 **A. (R)-3-Chloro-1-methylcyclohexene**

The C=C carbons are C1 and C2 (C1 has the CH_3), Cl on C3 → 3-chloro-1-methylcyclohexene. At C3: Cl > C2 (=CH) > C4 > H; Cl is wedged (H away) → clockwise → **R**.

Q22 **A. 3-Bromo-2-methylpentane**

Longest chain = 5 C. Numbering from the right gives {2,3} (from the left {3,4}) → **3-bromo-2-methylpentane**.

Q23 **D. 4-Bromocyclohexene**

The C=C must be C1–C2; going around the shorter way puts Br on C4 → **4-bromocyclohexene**.

Q24 **A. (1S,2S)-2-Chlorocyclohexan-1-ol**

OH (suffix -ol) = C1, Cl on C2, trans. C1: OH > C2 > C6 > H, H toward you → **S**. C2: Cl > C1 > C3 > H, H away → **S**.

Q25 **C. 4-Iodo-2-methylhexane**

6-carbon chain. From the left: {2,4}; from the right: {3,5} → use {2,4}: **4-iodo-2-methylhexane** (iodo before methyl alphabetically).

Q26 **A. 1-Bromo-2-ethylcyclopentane**

Both substituents get {1,2}; when there is a tie, the first in the alphabet (bromo) gets C1 → **1-bromo-2-ethylcyclopentane**.

12.4 Preparation of Alkyl Halides

Q27 **C. Addition of HX or X_2 to alkenes, and substitution of the OH of alcohols (HX, $SOCl_2$, PBr_3)**

(i) Alkene + HX → alkyl halide (Markovnikov); alkene + X_2 → vicinal dihalide. (ii) R–OH + HX → R–X + H_2O; R–OH + $SOCl_2$ → R–Cl + SO_2 + HCl; R–OH + PBr_3 → R–Br.

Q28 **C. $CH_3CHBrCH_3$ (2-bromopropane) — Markovnikov**

H goes to the carbon with more H atoms; Br goes to the more substituted carbon (more stable carbocation).

Q29 **B. $BrCH_2CH_2Br$ (1,2-dibromoethane, a vicinal dihalide)**

X_2 adds one X to each carbon of the C=C (anti addition).

Q30 **C. 1-Iodo-1-methylcyclohexane (structure B)**

HI adds by Markovnikov's rule: H goes to the =CH carbon, I goes to the carbon carrying the CH_3 (3° carbocation) → I and CH_3 on the same carbon.

Q31 **C. (Chloromethyl)cyclohexane**

$SOCl_2$ replaces the OH of an alcohol with Cl: $C_6H_{11}CH_2OH$ → $C_6H_{11}CH_2Cl + SO_2 + HCl$.

12.5 Reactions: Substitution & Elimination

Q32 **B. Nucleophilic substitution and elimination**

Substitution: $Nu^- + R–X$ → $R–Nu + X^-$ (e.g. $CH_3Br + OH^-$ → $CH_3OH + Br^-$). **Elimination (dehydrohalogenation):** a base removes H and X → alkene (e.g. $(CH_3)_3CBr + OH^-$ → $(CH_3)_2C=CH_2 + H_2O + Br^-$).

Q33 **C. An alkene (H and X are removed from neighbouring carbons)**

Base + H–C–C–X → C=C + H–Base + X^- (dehydrohalogenation).

Q34 **A. An electron-rich species (anion or neutral molecule with a lone pair) that donates an electron pair to an electron-poor carbon**

Nucleophile = "nucleus-loving"; a Lewis base. It attacks the δ+ carbon of the C–X bond and forms a new bond.

Q35 **A. OH^-, RO^- (alkoxide), H_2O, ROH, $RCOO^-$ (carboxylate)**

The nucleophilic atom is O.

Q36 **D. NH_3, RNH_2, R_2NH, R_3N, N_3^- (azide)**

Ammonia and amines attack with the N lone pair.

Q37 **B. HS^-, RS^- (thiolate), H_2S, RSH**

Sulphur nucleophiles are very good nucleophiles (large, polarizable).

Q38 **D. CN^- (cyanide) and $RC{\equiv}C^-$ (acetylide)**

Carbon nucleophiles make a new C–C bond: R–X + CN^- → R–CN; R–X + R'C≡C^- → R–C≡CR'.

Q39 **C. I^-, Br^-, Cl^- (halide ions)**

Halide ions can replace another halogen, e.g. R–Cl + I^- → R–I + Cl^-.

12.6 Nucleophilic Substitution Practice

Q40 **C. 2° alkyl halide (2-bromopropane); nucleophile = ethoxide ($C_2H_5O^-$); solvent = ethanol**

$CH_3CHBrCH_3$ = 2-bromopropane (Br on a carbon bonded to 2 carbons → 2°). The nucleophile is $C_2H_5O^-$ (Na^+ is only the counter-ion). Solvent: C_2H_5OH.

Q41 **B. Leaving group = Br^-; product = $C_2H_5OCH(CH_3)_2$ (2-ethoxypropane, an ether); by-product = NaBr**

Br^- leaves, ethoxide bonds to the carbon → ethyl isopropyl ether; Na^+ + Br^- = NaBr.

Q42 **B. $(CH_3)_2CHCH_2–O–CH_2CH_3$ (ethyl isobutyl ether) + NaBr**

Alkoxide (O nucleophile) + 1° alkyl halide → ether (Williamson ether synthesis).

Q43 **A. Ethyl cyclobutanecarboxylate (an ester) + KI**

The carboxylate ($RCOO^-$) is an O nucleophile; it replaces I^- → RCOO–CH_2CH_3 (ester).

Q44 **B. Cyclobutanethiol (cyclobutyl–SH) + KBr**

HS^- (S nucleophile) replaces Br^- → thiol R–SH.

Q45 **D. Cyclobutanecarbonitrile (cyclobutyl–C≡N) + NaBr**

CN^- (C nucleophile) replaces Br^- → nitrile R–CN (one more carbon).

Q46 **A. $CH_3CH_2C{\equiv}CCH_3$ (2-pentyne) + NaBr**

Acetylide (C nucleophile) + CH_3Br → the CH_3 joins the alkyne carbon → internal alkyne.

Q47 **A. $(CH_3)_2N^+H–CH_2CH_3\ I^-$ (ammonium salt), then with base → $(CH_3)_2N–CH_2CH_3$ (a 3° amine)**

The 2° amine (N nucleophile) attacks CH_3I → ammonium iodide salt. A base removes the H^+ from N → N,N-dimethylethylamine (3° amine).

Q48 **B. $CH_3(CH_2)_{13}N^+(CH_3)_3\ I^-$ — a quaternary ammonium salt**

A 3° amine + CH_3I → the N gets a fourth carbon and a + charge → quaternary ammonium iodide (no H on N, so it stays a salt).

Q49 **C. $CH_3C{\equiv}CCH_3$, 2-butyne (structure C)**

NaH removes the terminal alkyne H → acetylide $CH_3C{\equiv}C^-Na^+$ (C nucleophile). It attacks CH_3I → $CH_3C{\equiv}C–CH_3$.

Q50 **D. $CH_3CH_2CH_2CH_2Br + CH_3O^-Na^+$**

Ether = alkoxide + alkyl halide. ($CH_3Br + CH_3CH_2CH_2CH_2O^-$ also works.)

Q51 **B. $CH_3CH_2CH_2Br + HC{\equiv}C^-Na^+$**

Acetylide anion + 1° alkyl halide → new C–C bond.

Q52 **C. $CH_3CH_2CH_2Br + NH_3$ (excess)**

NH_3 (N nucleophile) replaces Br → 1° amine (excess NH_3 prevents further alkylation).

Q53 **B. $CH_3CH_2CH_2Br + CH_3CH_2CH_2S^-Na^+$**

Thiolate (RS^-) + alkyl halide → sulfide (thioether) R–S–R.

12.7 Safety of Organohalogen Compounds

Q54 **C. No — e.g. CCl_4 and $CHCl_3$ are toxic and can cause liver damage**

Some are useful drugs (chloramphenicol, halothane, thyroxine), but many are toxic: CCl_4 and $CHCl_3$ damage the liver (and some are carcinogenic).